Solution, Exam: FM1015 Modelling of Dynamic Systems

Date: Friday, November 29, 2024

Form of exam: This is a *Digital Exam* of type *Multiple* Choice. Each exam *question* (denoted "problems" in the text below) is followed by *4 possible answers* ("choice" in the tables related to questions). *Some* details about your answers and how they will be assessed:

- A problem gives full score if a correct and no incorrect choices are given, zero score otherwise.
- All problems are given the same weight.
- Most problems are posed so that it is necessary to do background development/calculations on paper to choose the correct answer (A–D).
- The background calculations shall not be submitted — the only submission in the Digital Exam are the choices (A–D) in each problem.
  - You need to insert the answers in the *attached* Excel sheet, and upload this Excel sheet which includes your answers, in the exam system [WISEflow].
  - The file you **upload must be in Excel file format**, i.e., *.xlsx format. (*Reason*: the files will be automatically parsed for grading. This parsing doesn't work if you upload a *.pdf file or any other file format.)
- A total score of 0-39% gives grade F (fail), score 40-49% gives grade E, score 50-59% gives grade D, score 60-79% gives grade C, score 80-89% gives grade B, and score 90-100% gives grade A.

Aids: No documents in printed or electronic form are allowed, no use of internet search allowed. Use of computer computation tools such as Python, Julia, or MATLAB are allowed. Necessary use of PC for using WISEflow, and the use of Excel for filling in answers is allowed.

This exam contains questions (denoted *problems*) in the *Problems* part. The Problems part consists of *20 problems*, each with 4 alternative choices (A–D), spread over 7 subsections.

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Comments to exam

1. On the top of the home page for the course in Canvas, there is a link ("Course description") to the legally binding web description of the course. On this web page, it is stated

   > "Examination support material. Written test: no documents in printed or electronic form are allowed, no use of internet search allowed. Use of computer computation tools such as Python, Julia, or MATLAB are allowed".

2. It is mentioned in the first lecture of the course (including the video) that this is the legally binding description of the course. An announcement was also filed several days before the exam in Canvas informing about these aids for the exam. It is unfortunate that there is a contradictory statement under the "Exam, etc." headline of the Canvas home page. This will be fixed for the next semester. However, sufficient information has been given so that this should not be a problem.

3. One candidate had not noticed this information, claimed to not have Python, MATLAB of Julia on her/his computer, and requested to be allowed to use a calculator. Because the exam administration is extremely strict wrt. following the legally binding web description, this request was not accepted. *However*, (i) in the solution below, it is demonstrated that the 3 problems involving numeric computations can easily be solved with pen/paper without using a calculator or computer in perhaps 5 minutes for each problem ("Hand calculation"). Furthermore, (ii) because it was necessary to use Excel to submit the answers, the numeric computations could easily be solved using Excel.

4. One candidate had not adhered to the required way of submitting answers. Instead of using the available Excel sheet and inserting the answers in that document, the candidate had instead *copied* the worksheet of the Excel file into a new Excel file, and written the answers in an incorrect column. I fail to see how the required procedure can be misunderstood. This break of procedure costed me 1 hour of extra work: the grading system didn't work, and it took me time to find which Excel file caused the problem + figure out how to solve it. In summary: there is a reason why you should follow the procedures (and everyone, except one, did).

5. The grade distribution on the exam seems fair:

   1×11 DataFrame

Row	A	B	C	D	E	F	abs
	Int64	Int64	Int64	Int64	Int64	Int64	Int64
1	6	9	24	10	3	3	3

1 Problems

In water electrolysis, water (H_2O) is split into hydrogen (H_2) and oxygen (O_2) by providing electric power. Because hydrogen and oxygen form an explosive mixture if mixed, a membrane is used to separate the generated hydrogen and oxygen.

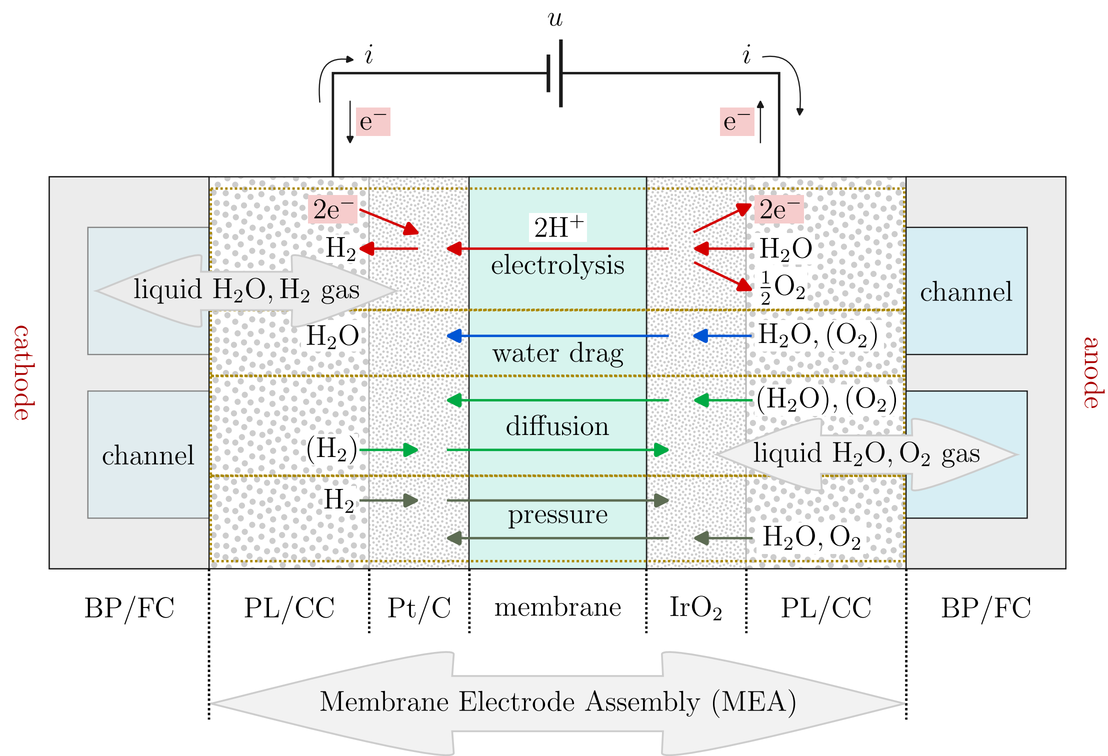

Figure 1: PEM Electrolysis Cell, with (left-to-right) indication of *cathode* side Bipolar Plate with Flow Channel (BP/FC), cathode side Porous Layer/Current Collector (PL/CC), cathode side Catalyst with Platinum on Carbon support (Pt/C). In the middle: solid membrane (Proton Exchange Membrane, PEM). Next, to the left: *anode* side Catalyst with Iridium Oxide catalyst (IrO_2), anode side PL/CC, and finally anode side BP/FC.

Hydrogen is stored at high pressure. The produced hydrogen can either be compressed up to the storage pressure (relatively expensive), or produced at the required pressure in the electrolyzer. This means that it may be advantageous to keep the hydrogen side of the electrolyzer close to the storage pressure.

In this exam, we consider a water electrolysis cell separating the produced hydrogen and oxygen by a Proton Exchange Membrane (PEM), see Fig. 1.

The surrounding infrastructure of a stack of PEM electrolyzer cells is depicted in Fig. 2.

Details of the PEM electrolysis system in Fig. 2 were studied in the compulsory student group project in 2024. With the group project in mind, consider the following problems — which have been organized under numbered sub-headings.

1.1 Stoichiometry

Consider an electrolyzer where water (H_2O) is available at the *anode* surface, and a voltage is set up across an acidic electrolyte membrane [PEM: proton exchange membrane, allowing for transport of *proton* (H^+) through the membrane]. This membrane separates the anode surface/volume from the cathode surface/volume, see Figs. 1 and 3.

In water electrolysis, electric power is added to drive the overall reaction

$$H_2O \rightarrow H_2 + \frac{1}{2}O_2. \tag{1}$$

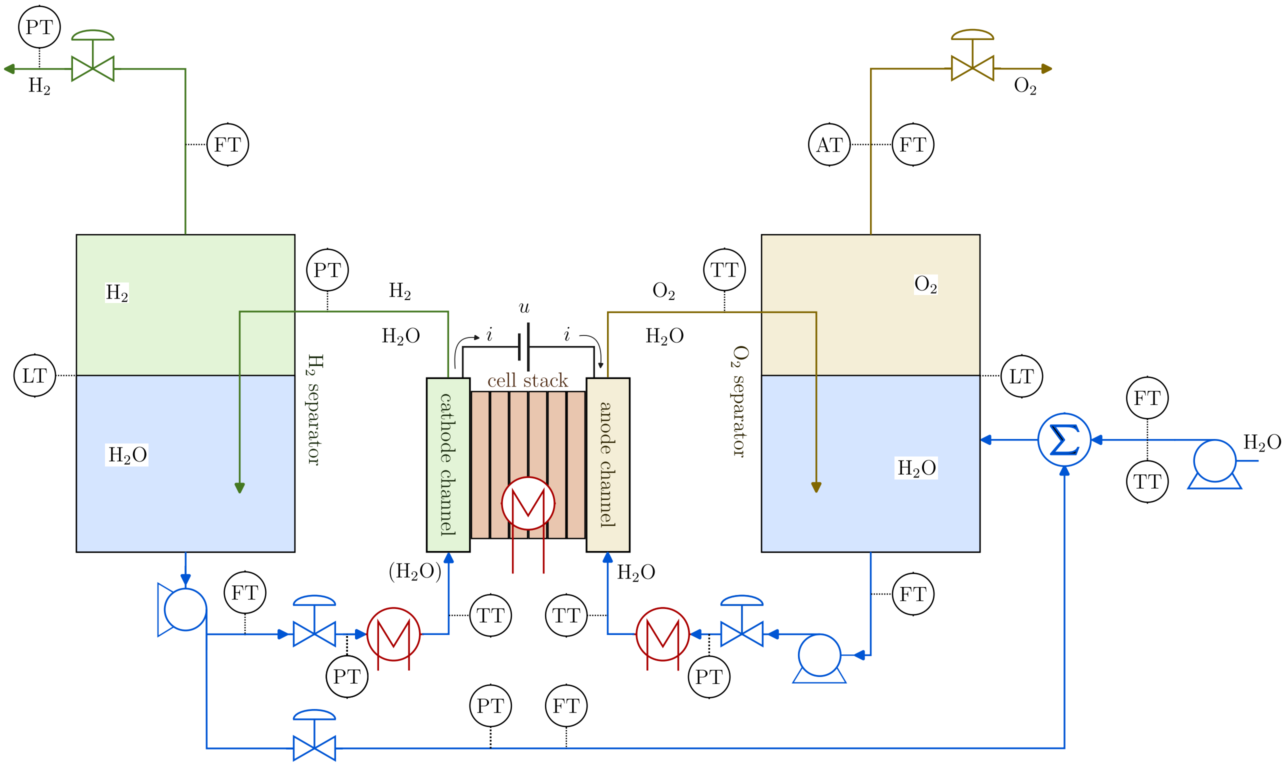

Figure 2: PEM Electrolysis System.

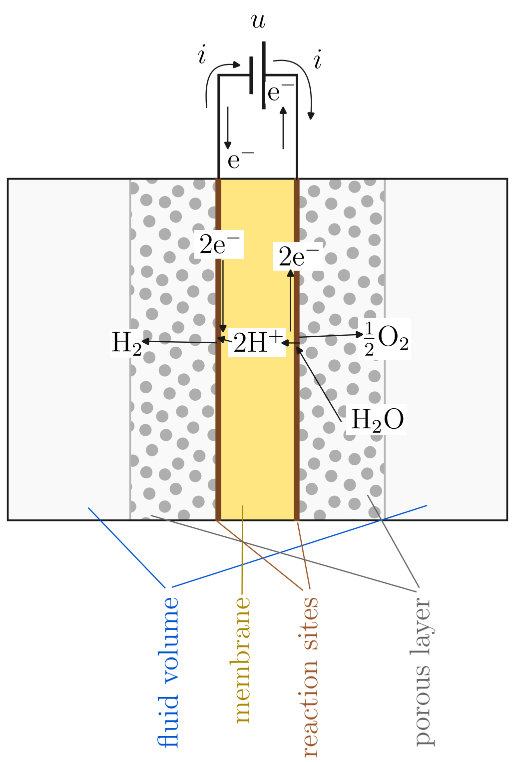

Figure 3: Principle of water electrolysis with acidic membrane.

With an acidic membrane, the catalyzed half-cell reaction at the anode surface produces oxygen [Oxygen Evolution Reaction, OER], while catalyzed half-cell reaction at the cathode surface produces hydrogen [Hydrogen Evolution Reaction, HER].

The overall half-cell reactions[1] are the anode (OER) half-cell reaction

$$\mathrm{H_2O} \rightarrow \frac{1}{2}\mathrm{O_2} + 2\mathrm{H^+} + 2\mathrm{e^-} \tag{2}$$

and the cathode (HER) half-cell reaction

$$2\mathrm{H^+} + 2\mathrm{e^-} \rightarrow \mathrm{H_2}. \tag{3}$$

This implies that oxygen is produced at the anode and hydrogen at the cathode. Proton (H^+) moves through the membrane, and water is consumed at the anode. Therefore, when using an acidic electrolyte (as in a PEM cell), water must be available on the anode side.

Problem 1. Reactions in Electrolyzer

Let the species be ordered according to the vector $(\mathrm{H_2O}, \mathrm{H_2}, \mathrm{O_2})$. It is of interest to formulate the *stoichiometric matrix* ν [possibly a row matrix/vector] for the overall reaction in Eq. 1, as well as the *positive stoichiometric parameter* for the *electron*, ν_{e^-} (Note: $\nu_{\mathrm{e}^-} > 0$).

	Problem 1: Which of the following pairs of ν and ν_{e^-} is ***incorrect***?	Choice
A:	$\nu = \left(-1, 1, \frac{1}{2}\right)$ and $\nu_{\mathrm{e}^-} = 2$	
B:	$\nu = \left(1, -1, -\frac{1}{2}\right)$ and $\nu_{\mathrm{e}^-} = 2$	
C:	$\nu = (-2, 2, 1)$ and $\nu_{\mathrm{e}^-} = 4$	
D:	$\nu = -(2, -2, -1)$ and $\nu_{\mathrm{e}^-} = 4$	

▲

Solution 1. Reactions in Electrolyzer

Let the species be ordered according to the vector $(\mathrm{H_2O}, \mathrm{H_2}, \mathrm{O_2})$. It is of interest to formulate the *stoichiometric matrix* ν for the overall reaction in Eq. 1, as well as the *positive stoichiometric parameter* for the *electron*, ν_{e^-} (Note: $\nu_{\mathrm{e}^-} > 0$).

In the case of an *acidic* electrolyte/membrane, the overall half-cell reactions are the anode (oxidation reaction, OER)

$$\mathrm{H_2O} \rightarrow \frac{1}{2}\mathrm{O_2} + 2\mathrm{H^+} + 2\mathrm{e^-} \tag{4}$$

and the overall half-cell reaction at the cathode (reduction reaction, HER)

$$2\mathrm{H^+} + 2\mathrm{e^-} \rightarrow \mathrm{H_2}. \tag{5}$$

This implies that oxygen is produced at the anode and hydrogen at the cathode. Proton (H^+) moves through the membrane, and water is consumed at the anode. Therefore, when using an acidic electrolyte, water must be supplied on the anode side.

[1]"overall half-cell reactions": the real reactions are much more complex, so the listed half-cell reactions are the overall reactions at the half-cells.

Combining the two half-cell reactions, proceed as follows: take the oxidation reaction and move all terms to the LHS[2] of the arrow, and take the reduction reaction and move all terms to the RHS[3] of the arrow. Adding the expressions, we get

$$1 \cdot \left(\mathrm{H_2O} - \frac{1}{2}\mathrm{O_2} - 2\mathrm{H^+} - 2\mathrm{e^-} \right) \to 1 \cdot \left(\mathrm{H_2} - 2\mathrm{H^+} - 2\mathrm{e^-} \right)$$

$$\Downarrow$$

$$\mathrm{H_2O} \to \mathrm{H_2} + \frac{1}{2}\mathrm{O_2} + \underbrace{2}_{\nu_{\mathrm{e^-}}=2} \cdot \left(\mathrm{e^-} - \mathrm{e^-}\right) + 2 \cdot \left(\mathrm{H^+} - \mathrm{H^+}\right)$$

$$\Downarrow$$

$$\mathrm{H_2O} \to \mathrm{H_2} + \frac{1}{2}\mathrm{O_2}.$$

Considering hydrogen and oxygen as products, and with species vector $\mathrm{S} = (\mathrm{H_2O}, \mathrm{H_2}, \mathrm{O_2})$, we find stoichiometric vector/matrix $\nu^{(1)}$ and (hidden) electron stoichiometric coefficient $\nu_{\mathrm{e^-}}$ to be

$$\begin{aligned} \nu^{(1)} &= \begin{pmatrix} -1 & 1 & \frac{1}{2} \end{pmatrix} \\ \nu_{\mathrm{e^-}} &= 2. \end{aligned}$$

But we could just has well have multiplied the entire reaction with a constant k, and would then have

$$\begin{aligned} \nu^{(k)} &= k \cdot \begin{pmatrix} -1 & 1 & \frac{1}{2} \end{pmatrix} \\ \nu_{\mathrm{e^-}} &= 2k; \end{aligned}$$

constant k should be positive to have a positive $\nu_{\mathrm{e^{-1}}}$. We would still have scaled stoichiometric matrix

$$\frac{\nu^{(1)}}{\nu_{\mathrm{e^-}}} = \frac{\nu^{(k)}}{\nu_{\mathrm{e^-}}} = \left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{4} \right)$$

— which is invariant wrt. choice of coefficients in the overall reactions. It follows that *Choice B* is the correct choice.

	Problem 1: Which of the following pairs of ν and $\nu_{\mathrm{e^-}}$ is ***incorrect***?	Choice
A:	$\nu = \left(-1, 1, \frac{1}{2}\right)$ and $\nu_{\mathrm{e^-}} = 2$	
B:	$\nu = \left(1, -1, -\frac{1}{2}\right)$ and $\nu_{\mathrm{e^-}} = 2$	×
C:	$\nu = (-2, 2, 1)$ and $\nu_{\mathrm{e^-}} = 4$	
D:	$\nu = -(2, -2, -1)$ and $\nu_{\mathrm{e^-}} = 4$	

▲

1.2 Electrolyzer voltages

The so-called *chemical potential* μ_j of species j in a mixture is given as molar Gibbs' energy $\tilde{G}_j^\circ$ of species j at standard state plus a term $RT \cdot \ln a_j$ where R is the gas constant, T is absolute temperature (i.e., in K), and a_j is the *activity*. Introducing $\mu_j \equiv \tilde{G}_j$, we have

$$\tilde{G}_j = \tilde{G}_j^\circ + RT \cdot \ln a_j. \tag{6}$$

[2]Left Hand Side
[3]Right Hand Side

For *ideal gas*,

$$a_j = \frac{p_j}{p^\circ} \tag{7}$$

where p_j is the partial pressure of species j, while p° is a standard state *scaling* pressure (normally either 1 atm or 1 bar). In the liquid case, for an *ideal solution*

$$a_j = x_j \tag{8}$$

where x_j is the mole fraction $x_j = n_j/n$ of the liquid. For pure liquid j, $x_j = 1$.

The molar Gibbs energy of *reaction* $\Delta_\mathrm{r}\tilde{G}$ (at arbitrary T, p_j) is given as

$$\Delta_\mathrm{r}\tilde{G} = \sum_j \nu_j \tilde{G}_j, \tag{9}$$

while the *standard* molar Gibbs energy of reaction (i.e., at arbitrary T and standard pressure p°) is denoted $\Delta_\mathrm{r}\tilde{G}^\circ$. The equilibrium voltage from the anode side (positive) to the cathode side (negative) is then

$$u_\mathrm{n} = \frac{\Delta_\mathrm{r}\tilde{G}}{F\nu_{\mathrm{e}^-}} \tag{10}$$

where F is *Faraday's number*.

Problem 2. Equilibrium voltage in Electrolyzer

The Nernst equilibrium voltage u_n can be expressed in different ways.

	Problem 2: Equilibrium voltage — which expression is ***incorrect***?	Choice
A:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ + RT\prod_j \ln a_j^{\nu_j}\right) / (F\nu_{\mathrm{e}^-})$	
B:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ + RT\sum_j \ln a_j^{\nu_j}\right) / (F\nu_{\mathrm{e}^-})$	
C:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ + RT\ln\left(\prod_j a_j^{\nu_j}\right)\right) / (F\nu_{\mathrm{e}^-})$	
D:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ/\nu_{\mathrm{e}^-} + RT\ln\left(\prod_j a_j^{\nu_j/\nu_{\mathrm{e}^-}}\right)\right) / F$	

▲

Solution 2. Equilibrium voltage in Electrolyzer

The Nernst voltage u_n can be expressed as

$$u_\mathrm{n} = \frac{\Delta_\mathrm{r}\tilde{G}}{F\nu_{\mathrm{e}^-}} = \frac{\sum_j \nu_j \tilde{G}_j}{F\nu_{\mathrm{e}^-}} = \frac{\sum_j \nu_j\left(\tilde{G}_j^\circ + RT\cdot\ln a_j\right)}{F\nu_{\mathrm{e}^-}}$$

$$\Downarrow$$

$$u_\mathrm{n} = \frac{\sum_j \nu_j \tilde{G}_j^\circ + \sum_j \nu_j RT\cdot\ln a_j}{F\nu_{\mathrm{e}^-}}.$$

Introducing

$$\Delta_\mathrm{r}\tilde{G}^\circ \triangleq \sum_j \nu_j \tilde{G}_j^\circ,$$

this gives

$$u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ} + \sum_j \nu_j RT \cdot \ln a_j}{F\nu_{\mathrm{e}^-}}$$
$$\Downarrow$$
$$u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ} + RT\sum_j \nu_j \cdot \ln a_j}{F\nu_{\mathrm{e}^-}}. \tag{11}$$

The expression in Eq. 11 is numerically the best to use. Two basic rules for logarithms are

$$a \ln x = \ln x^a \tag{12}$$
$$\sum_j \ln x_j = \ln\left(\prod_j x_j\right). \tag{13}$$

Application of Eq. 12 to Eq. 11 allows us to develop Choice B:

$$u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ} + RT\sum_j \ln a_j^{\nu_j}}{F\nu_{\mathrm{e}^-}}. \tag{14}$$

Alternatively, by a slight re-write of Eq. 14 in combination with Eq. 12, we can develop Choice C:

$$u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ} + RT\sum_j \ln a_j^{\nu_j}}{F\nu_{\mathrm{e}^-}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ} + RT\ln\left(\prod_j a_j^{\nu_j}\right)}{F\nu_{\mathrm{e}^-}}. \tag{15}$$

Finally, by a slight re-write of Choice B in Eq. 14, together with Eq. 12, we can develop Choice D,

$$u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ} + RT\sum_j \nu_j \cdot \ln a_j}{F\nu_{\mathrm{e}^-}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ}/\nu_{\mathrm{e}^-} + RT\sum_j \frac{\nu_j}{\nu_{\mathrm{e}^-}} \cdot \ln a_j}{F}$$
$$\Downarrow$$
$$u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ}/\nu_{\mathrm{e}^-} + RT\sum_j \ln a_j^{\nu_j/\nu_{\mathrm{e}^-}}}{F} = \frac{\Delta_{\mathrm{r}}\tilde{G}^{\circ}/\nu_{\mathrm{e}^-} + RT\ln\left(\prod_j a_j^{\nu_j/\nu_{\mathrm{e}^-}}\right)}{F}. \tag{16}$$

We see that Choice A is ***incorrect*** because (and thus the correct choice)

$$\sum_j \ln a_j^{\nu_j} \neq \prod_j \ln a_j^{\nu_j}.$$

A final comment: why is the formulation in Eq. 11 superior to the other formulations? The reason is simple: in the case that ν_j is a non-integer, say $\nu_j = \frac{1}{2}$, in Eq. 11 we simply take the logarithm of a_j, followed by multiplying the result with ν_j; multiplication is a very efficient operation on computers.

In summary: *Choice A* is the correct choice.

In Eqs. 14–16, we replace the simple multiplication operation with potentially raising a_j to the power of a non-integer, e.g., $a_j^{\nu_j}$ or $a_j^{\nu_j/\nu_{\mathrm{e}^-}}$, which is an inefficient operation, before taking the logarithm. In fact, for the particular case that $\nu_j = \frac{1}{2}$, it would actually be more efficient to compute $\sqrt{a_j}$ instead of $a_j^{1/2}$ because doing $\sqrt{a_j}$ would use a built-in, relatively efficient algorithm for taking the square root, while $a_j^{1/2}$ would use a generic, relatively inefficient algorithm to raise a number to a power.

	Problem 2: Equilibrium voltage — which expression is ***incorrect***?	Choice
A:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ + RT\prod_j \ln a_j^{\nu_j}\right) / (F\nu_{\mathrm{e}^-})$	×
B:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ + RT\sum_j \ln a_j^{\nu_j}\right) / (F\nu_{\mathrm{e}^-})$	
C:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ + RT\ln\left(\prod_j a_j^{\nu_j}\right)\right) / (F\nu_{\mathrm{e}^-})$	
D:	$u_\mathrm{n} = \left(\Delta_\mathrm{r}\tilde{G}^\circ/\nu_{\mathrm{e}^-} + RT\ln\left(\prod_j a_j^{\nu_j/\nu_{\mathrm{e}^-}}\right)\right) / F$	

▲

Although various formulation are possible for u_n, it is numerically advantageous to compute u_n by combining Eqs. 9 and 6 into equilibrium voltage u_n (Nernst voltage) as

$$u_\mathrm{n} = \underbrace{\frac{\sum_j \nu_j \tilde{G}_j^\circ}{F\nu_{\mathrm{e}^-}}}_{=\frac{\Delta_\mathrm{r}\tilde{G}^\circ}{F\nu_{\mathrm{e}^-}}=u_\circ} + \underbrace{\frac{RT}{F\nu_{\mathrm{e}^-}} \cdot \sum_j \nu_j \ln a_j}_{=u_\mathrm{p}}, \tag{17}$$

where $u_\circ$ represents the equilibrium voltage at standard state (the actual temperature T, with standard state pressure $p^\circ = 1\,\mathrm{bar}$), while u_p represents the pressure/activity dependent change in equilibrium voltage when $p \neq p^\circ$. Here, molar Gibbs energy at standard state, $\tilde{G}_j^\circ$, can to a good approximation be found from

$$\tilde{G}_j^\circ = \Delta_\mathrm{f}\tilde{G}_j^\circ - \Delta_\mathrm{f}\tilde{S}_j^\circ\,(T - T^\circ) + \tilde{c}_{\mathrm{p},j}\,(T - T^\circ) - T\cdot\tilde{c}_{\mathrm{p},j}\ln\frac{T}{T^\circ} \tag{18}$$

where for each species j, values for molar Gibbs energy of formation $\Delta_\mathrm{f}\tilde{G}_j^\circ$, molar entropy of formation $\Delta_\mathrm{f}\tilde{S}_j^\circ$, and molar heat capacity at constant pressure $\tilde{c}_{\mathrm{p},j}$ can be found in physical tables. For the overall equilibrium reaction $H_2O \rightleftharpoons H_2 + \frac{1}{2}O_2$, this leads to the variation in u_n with temperature as in Fig. 4.

Problem 3. Equilibrium voltage with pressure variation

In Fig. 4, the case $(p_{H_2}, p_{O_2}) = (1,1)\,\mathrm{bar}$ (solid line) represents $u_\circ$. From Fig. 4, the following assessments of u_n can be given.

	Problem 3: Equilibrium voltage — which statement is ***correct***?	Choice
A:	With increasing H_2 pressure and O_2 pressure, the equilibrium voltage decreases.	
B:	The pressure dependent change in equilibrium voltage u_n is always a disadvantage.	
C:	With increasing H_2 pressure and O_2 pressure, the equilibrium voltage increases.	
D:	The pressure dependent change in equilibrium voltage u_n is always an advantage.	

▲

Solution 3. Equilibrium voltage with pressure variation

In Fig. 4, the case $(p_{H_2}, p_{O_2}) = (1,1)\,\mathrm{bar}$ (solid line) represents $u_\circ$. From Fig. 4, the following assessments of u_n can be given.

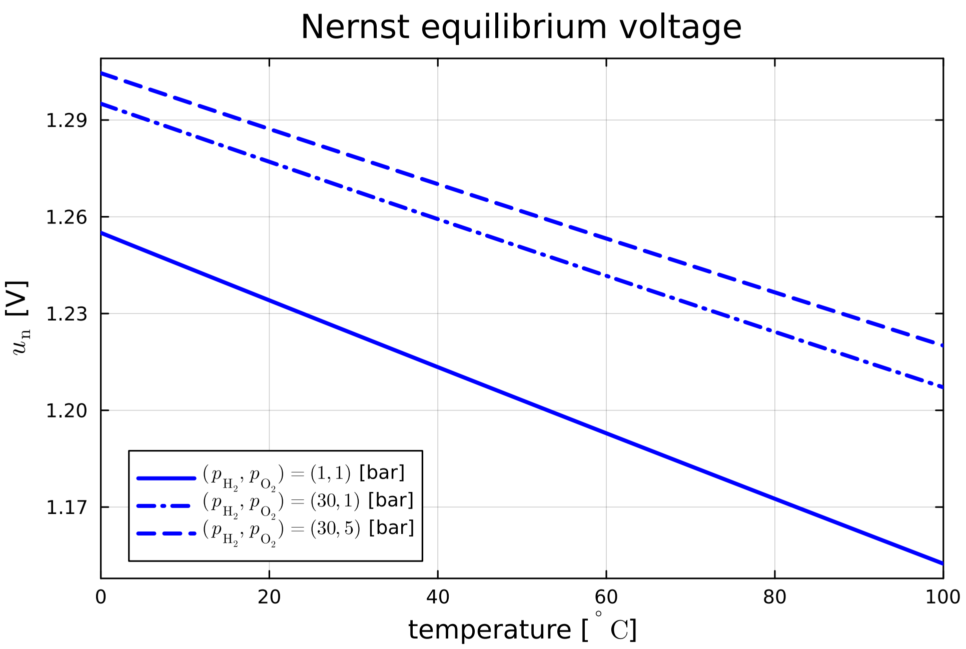

Figure 4: Nernst voltage u_n as a function of temperature for three combinations of H_2 and O_2 pressures.

- With increasing pressures H_2 and O_2, the equilibrium voltage u_n **increases**, so *Choice C* is a correct statement. This also means that Choice A is wrong.
- The pressure dependent change in equilibrium voltage u_n is a disadvantage in that at higher pressure, more electric power is needed to drive the reaction (seemingly, Choice B might be correct and Choice D is wrong). However, the advantages of higher pressure are (i) the hydrogen product may be produced at a higher pressure thus requiring less power to compress hydrogen to storage pressure, and (ii) in Problem 9, we will see that a higher oxygen pressure in fact reduces the loss of water to the atmosphere — it is not clear how big an advantage this is, though. In summary, it is not possible to claim that the pressure dependence is always a disadvantage, or always an advantage; there are advantages and disadvantages.

	Problem 3: Equilibrium voltage — which statement is ***correct***?	Choice
A:	With increasing H_2 pressure and O_2 pressure, the equilibrium voltage decreases.	
B:	The pressure dependent change in equilibrium voltage u_n is always a disadvantage.	
C:	With increasing H_2 pressure and O_2 pressure, the equilibrium voltage increases.	×
D:	The pressure dependent change in equilibrium voltage u_n is always an advantage.	

▲

When current i_c starts to flow through the cell, additional voltage drops occur, such as activation voltages (u_a) and membrane resistance voltage (u_r). These additional voltages are functions of temperature, but also of the current *density* i'', defined via Eq. 19,

$$i_c = A_c i'' \tag{19}$$

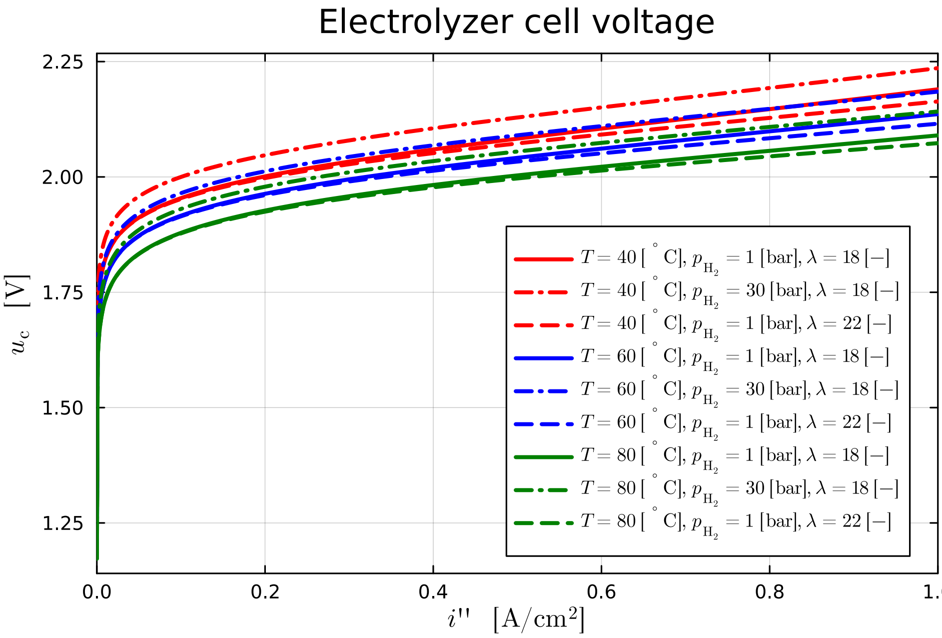

Figure 5: Electrolyzer cell voltage operating at $T = 40\,°\mathrm{C}$ [red], $T = 60\,°\mathrm{C}$ [blue], and $T = 80\,°\mathrm{C}$ [green] at $p_{\mathrm{H}_2} = 1\,\mathrm{bar}, \lambda = 18$ [solid], $p_{\mathrm{H}_2} = 30\,\mathrm{bar}, \lambda = 18$ [dash-dot], and $p_{\mathrm{H}_2} = 1\,\mathrm{bar}, \lambda = 22$ [dash]; $p_{\mathrm{O}_2} = 1\,\mathrm{bar}$.

where A_c is the membrane area of the cell. The resulting cell voltage u_c is

$$u_\mathrm{c} = u_\mathrm{n} + u_\mathrm{a} + u_\mathrm{r}, \tag{20}$$

as in Fig. 5.

Note: with modern catalysts and thinner membranes, u_c is typically down to 1.8 V or less $i'' = 2\,\mathrm{A/cm^2}$.

Problem 4. Cell voltage

In the group project, the case $T = 60\,°\mathrm{C}$ with pressures $(p_{\mathrm{H}_2}, p_{\mathrm{O}_2}) = (30, 1)\,\mathrm{bar}$ was considered. Assume $\lambda = 18$, and give estimates of u_n and u_c for these conditions at current density $i'' = 1\,\mathrm{A/cm^2}$.

	Problem 4: Equilibrium and cell voltages — which statement is ***correct***?	Choice
A:	$u_\mathrm{n} \approx 1.20\,\mathrm{V}$, $u_\mathrm{c} \approx 2.15\,\mathrm{V}$	
B:	$u_\mathrm{n} \approx 1.24\,\mathrm{V}$, $u_\mathrm{c} \approx 2.15\,\mathrm{V}$	
C:	$u_\mathrm{n} \approx 1.24\,\mathrm{V}$, $u_\mathrm{c} \approx 2.00\,\mathrm{V}$	
D:	$u_\mathrm{n} \approx 1.20\,\mathrm{V}$, $u_\mathrm{c} \approx 2.00\,\mathrm{V}$	

▲

Solution 4. Cell voltage

In the group project, the case $T = 60\,°\mathrm{C}$ with pressures $(p_{\mathrm{H}_2}, p_{\mathrm{O}_2}) = (30, 1)\,\mathrm{bar}$ was considered. Assume $\lambda = 18$, and give estimates of u_n and u_c for these conditions at current density $i'' = 1\,\mathrm{A/cm^2}$.

From Fig. 4, we see that the dash-dot line ($(p_{\mathrm{H}_2}, p_{\mathrm{O}_2}) = (30, 1)\,\mathrm{bar}$) has an equilibrium voltage of approximately $u_\mathrm{n} \approx 1.24\,\mathrm{V}$ at $T = 60\,°\mathrm{C}$.

From Fig. 5, which uses $p_{O_2} = 1\,\mathrm{bar}$, at $T = 60\,°\mathrm{C}$ (blue curves) and pressure-sorption combination $p_{H_2} = 30\,\mathrm{bar}$, $\lambda = 18$ (dash-dot curve), we see that $u_c \approx 2.15\,\mathrm{V}$ at $i'' = 1\,\mathrm{A/cm^2}$. In other words, *Choice B* is correct.

[Observe that u_c is not shown at $i'' = 2\,\mathrm{A/cm^2}$ — which is referred to later in the exam. In the later problems with $i'' = 2\,\mathrm{A/cm^2}$, $u_c\,(i'' = 2)$ is estimated to $u_c\,(i'' = 2) \approx 2\,\mathrm{V}$.]

	Problem 4: Equilibrium and cell voltages — which statement is ***correct***?	Choice
A:	$u_n \approx 1.20\,\mathrm{V}$, $u_c \approx 2.15\,\mathrm{V}$	
B:	$u_n \approx 1.24\,\mathrm{V}$, $u_c \approx 2.15\,\mathrm{V}$	×
C:	$u_n \approx 1.24\,\mathrm{V}$, $u_c \approx 2.00\,\mathrm{V}$	
D:	$u_n \approx 1.20\,\mathrm{V}$, $u_c \approx 2.00\,\mathrm{V}$	

▲

1.3 Electrolyzer stack design

Electric power consumption ("work" flow) in an electrolyzer stack is $\dot{W}_s = u_s i_s$ (unit: W) where u_s (unit: V) is the stack voltage, while i_s (unit: A) is the stack current. A stack consists of N_c cells. PEM electrolyzer cells are normally organized electrically in series (bipolar cells), stack voltage u_s is then

$$u_s = N_c u_c \tag{21}$$

where u_c is the cell voltage, and stack current i_s is

$$i_s = i_c \tag{22}$$

where i_c is the current through a cell.

Generation of species j can be expressed as

$$\dot{n}_j^g = \nu_j \dot{n}_g \tag{23}$$

where the *generation rate* $\dot{n}_g$ is

$$\dot{n}_g = \frac{N_c A_c}{\nu_{e^-} F} i''. \tag{24}$$

The corresponding mass generation rate is

$$\dot{m}_j^g = M_j \dot{n}_j^g = M_j \frac{\nu_j N_c A_c}{\nu_{e^-} F} i'', \tag{25}$$

where a negative value for $\dot{n}_j^g$ and $\dot{m}_j^g$ indicates *consumption*.

In the design of a stack, assume that maximally allowed current density is i''_{max}, which leads to a corresponding maximal cell voltage $u_c\,(i''_{\mathrm{max}})$. With N_c cells in the stack, the maximal electric power $\dot{W}_s^{\mathrm{max}}$ added to the stack is

$$\dot{W}_s^{\mathrm{max}} = u_s^{\mathrm{max}} i_s^{\mathrm{max}} = N_c u_c\,(i''_{\mathrm{max}})\, i_c^{\mathrm{max}} = N_c A_c i''_{\mathrm{max}} \cdot u_c\,(i''_{\mathrm{max}})\,. \tag{26}$$

By combining Eqs. 25 and 26, the corresponding maximal hydrogen production is

$$\dot{m}_{H_2}^{\mathrm{max}} = \frac{\nu_{H_2} M_{H_2}}{\nu_{e^-} F} N_c A_c i''_{\mathrm{max}}. \tag{27}$$

Solving Eq. 27 for $N_c A_c i''_{max}$ and inserting the resulting expression into Eq. 26, it follows that with a specified hydrogen production rate, the required electric power delivery to the stack is

$$\dot{W}_s^{max} = \frac{\nu_{e^-} F \cdot u_c\left(i''_{max}\right)}{\nu_{H_2} M_{H_2}} \dot{m}_{H_2}^{max}. \tag{28}$$

From Eq. 26, and with $\dot{W}_s^{max}$ as in Eq. 28, we can find the total membrane area A_s of the stack as

$$A_s = N_c A_c = \frac{\dot{W}_s^{max}}{i''_{max} u_c\left(i''_{max}\right)}. \tag{29}$$

Problem 5. Stack design I

In the project group, the following data were used

$$\begin{aligned} i''_{max} &= 2\,\mathrm{A/cm^2} \\ M_{H_2} &= 2\,\mathrm{g/mol} = 2 \cdot 10^{-3}\,\mathrm{kg/mol} \\ F &= 9.649 \cdot 10^4\,\mathrm{C/mol} \end{aligned}$$

For simplicity, set $u_c\left(i''_{max}\right) = 2.5\,\mathrm{V}$ (note: Fig. 5 only displays u_c for $i'' \in [0, 1]$). With required production $\dot{m}_{H_2}^{max} = 24\,\mathrm{kg/h} = \frac{24}{3600}\,\mathrm{kg/s}$, find the numeric value of the stack membrane area $A_s = N_c A_c$.

	Problem 5: With the given data, the stack membrane area A_s is — which value is **correct?**	Choice
A:	$A_s \approx 160\,\mathrm{m^2}$	
B:	$A_s \approx 1.6 \cdot 10^6\,\mathrm{m^2}$	
C:	$A_s \approx 3.2 \cdot 10^5\,\mathrm{m^2}$	
D:	$A_s \approx 32\,\mathrm{m^2}$	

▲

Solution 5. Stack design I

In the project group, the following data were used

$$\begin{aligned} i''_{max} &= 2\,\mathrm{A/cm^2} \\ M_{H_2} &= 2\,\mathrm{g/mol} = 2 \cdot 10^{-3}\,\mathrm{kg/mol} \\ F &= 9.649 \cdot 10^4\,\mathrm{C/mol} \end{aligned}$$

For simplicity, set $u_c\left(i''_{max}\right) = 2.5\,\mathrm{V}$ (note: Fig. 5 only displays u_c for $i'' \in [0, 1]$). With required production $\dot{m}_{H_2}^{max} = 24\,\mathrm{kg/h} = \frac{24}{3600}\,\mathrm{kg/s}$, find the numeric value of the stack membrane area $A_s = N_c A_c$.

Using Eq. 28, we find that

$$\dot{W}_s^{max} = \frac{\nu_{e^-} F \cdot u_c\left(i''_{max}\right)}{\nu_{H_2} M_{H_2}} \dot{m}_{H_2}^{max} = \frac{2 \cdot 9.649 \cdot 10^4 \cdot 2.5}{1 \cdot 2 \cdot 10^{-3}} \cdot \frac{24}{3600} \approx 1.6 \cdot 10^6\,\mathrm{W}.$$

Inserting this value into Eq. 29 gives

$$A_s = \frac{\dot{W}_s^{max}}{i''_{max} u_c\left(i''_{max}\right)} = \frac{1.6 \cdot 10^6}{2 \cdot 2.5} \approx 3.2 \cdot 10^5\,\mathrm{cm^2} = 3.2 \cdot 10^5 \cdot 10^{-4}\,\mathrm{m^2} = 32\,\mathrm{m^2}.$$

It follows that *Choice D* is correct.

	Problem 5: With the given data, the stack membrane area A_s is — which value is **correct?**	Choice
A:	$A_s \approx 160\,\mathrm{m}^2$	
B:	$A_s \approx 1.6 \cdot 10^6\,\mathrm{m}^2$	
C:	$A_s \approx 3.2 \cdot 10^5\,\mathrm{m}^2$	
D:	$A_s \approx 32\,\mathrm{m}^2$	×

Hand calculation: If no computer tool is used, it is possible to estimate the stack area A_s as follows:

$$\dot{W}_s^{\max} = \frac{\nu_{e^-} F \cdot u_c\left(i''_{\max}\right)}{\nu_{H_2} M_{H_2}} \dot{m}_{H_2}^{\max} = \frac{2 \cdot 9.649 \cdot 10^4 \cdot 2.5}{1 \cdot 2 \cdot 10^{-3}} \cdot \frac{24}{3600}$$

$$\Downarrow$$

$$\dot{W}_s^{\max} \approx \frac{2 \cdot 10 \cdot 10^4 \cdot 2.5}{1 \cdot 2 \cdot 10^{-3}} \cdot \frac{30}{5000} \approx \frac{10^5}{10^{-3}} \cdot 2.5 \cdot \frac{30}{0.5 \cdot 10^4} = 10^8 \cdot \frac{2.5}{0.5} \cdot \frac{30}{10^4}$$

$$\Downarrow$$

$$\dot{W}_s^{\max} \approx 10^4 \cdot 5 \cdot 30 = 1.5 \cdot 10^6\,\mathrm{W}.$$

This is not exact, but close enough. Next, we find the area

$$A_s = \frac{\dot{W}_s^{\max}}{i''_{\max} u_c\left(i''_{\max}\right)} \approx \frac{1.5 \cdot 10^6}{2 \cdot 2.5} = 3 \cdot 10^5\,\mathrm{cm}^2 = 3 \cdot 10^5 \cdot 10^{-4}\,\mathrm{m}^2 = 30\,\mathrm{m}^2.$$

This is still an approximation, but more than accurate enough to find the correct alternative.

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With a required stack membrane area A_s, choice of cell membrane area A_c provides the required number of cells N_c. In the group project, the membrane area was set to $A_c = 160\,\mathrm{cm}^2$, which lead to $N_c \approx 2000$.

In industrial design, typically A_c is in the range of 400–500 cm^2. Furthermore, normally N_c is rarely larger than $N_c = 200$. If the realistic values for N_c and A_c lead to a stack membrane area $A_s = N_c A_c$ that is smaller than what is required by Eq. 29, a possible solution is to connect several stacks in parallel.

Problem 6. Stack design II

With $N_c = 200$, the stack voltage is $u_s = N_c u_c$ or with $u_c = u_c\left(i''_{\max}\right) = 2.5\,\mathrm{V}$, around 500 V. Assuming $A_c = 400\,\mathrm{cm}^2$, the total area for the stack is $N_s = N_c A_c = 8 \cdot 10^4\,\mathrm{cm}^2$, which is smaller than the required number N_s found in Problem 5.

To achieve the required production of $\dot{m}_{H_2}^{\max}$, we therefore need to operate with more than one stack in parallel. It is also of interest to consider the maximal total current into the stack system.

	Problem 6: The minimal number of stacks ($A_c = 400\,\mathrm{cm}^2$, $N_c = 200$), and the resulting total current for the system are — which answer is **correct?**	Choice
A:	We need 3 stacks, and the total current is $i = 800\,\mathrm{A}$.	
B:	We need 3 stacks, and the total current is $i = 8\,\mathrm{kA}$.	
C:	We need 4 stacks, and the total current is $i = 320\,\mathrm{A}$.	
D:	We need 4 stacks, and the total current is $i = 3.2\,\mathrm{kA}$.	

▲

Solution 6. Stack design II

With $N_\mathrm{c} = 200$, the stack voltage is $u_\mathrm{s} = N_\mathrm{c} u_\mathrm{c}$ or with $u_\mathrm{c} = u_\mathrm{c}(i''_\mathrm{max}) = 2.5\,\mathrm{V}$, around $500\,\mathrm{V}$. Assuming $A_\mathrm{c} = 400\,\mathrm{cm}^2$, the total area for the stack is $N_\mathrm{s} = N_\mathrm{c} A_\mathrm{c} = 8 \cdot 10^4\,\mathrm{cm}^2$, which is smaller than the required number N_s found in Problem 5.

To achieve the required production of $\dot{m}^\mathrm{max}_{\mathrm{H}_2}$, we therefore need to operate with more than one stack in parallel. It is also of interest to consider the maximal total current into the stack system.

The maximal stack area with $A_\mathrm{c} = 400\,\mathrm{cm}^2$ and maximal value of cells in the stack $N^\mathrm{max}_\mathrm{c} = 200$, is $A^\mathrm{max}_\mathrm{s} = 400 \cdot 200 = 8 \cdot 10^4\,\mathrm{cm}^2 = 8\,\mathrm{m}^2$. This means that with a production requiring $A_\mathrm{s} = 32\,\mathrm{m}^2$, we need to operate $32/8 = 4$ stacks in parallel. For each stack, the current is $i_\mathrm{s} = i_\mathrm{c} = A_\mathrm{c} \cdot i''_\mathrm{max} = 400 \cdot 2 = 800\,\mathrm{A}$. Because the four stacks are organized in parallel, we need to multiply the total current with 4 , i.e., the total current is $i = 800 \cdot 4 = 3200\,\mathrm{A} = 3.2\,\mathrm{kA}$. In other words: *Choice D* is correct.

	Problem 6: The minimal number of stacks ($A_\mathrm{c} = 400\,\mathrm{cm}^2$, $N_\mathrm{c} = 200$), and the resulting total current for the system are — which answer is **correct?**	Choice
A:	We need 3 stacks, and the total current is $i = 800\,\mathrm{A}$.	
B:	We need 3 stacks, and the total current is $i = 8\,\mathrm{kA}$.	
C:	We need 4 stacks, and the total current is $i = 320\,\mathrm{A}$.	
D:	We need 4 stacks, and the total current is $i = 3.2\,\mathrm{kA}$.	×

Hand calculation: Again, we have $A^\mathrm{max}_\mathrm{s} = 400 \cdot 200 = 8 \cdot 10^4\,\mathrm{cm}^2 = 8\,\mathrm{m}^2$. With $A_\mathrm{s} \approx 30\,\mathrm{m}^2$ from Problem 5, this is quite close to four times $A^\mathrm{max}_\mathrm{s}$, so it is perhaps natural to guess 4 stacks, which limits the choice to options C or D. With maximal current density $i''_\mathrm{max} = 2\,\mathrm{A/cm}^2$ and membrane area $A_\mathrm{c} = 400\,\mathrm{cm}^2$, this gives $i = 800\,\mathrm{A}$ for one stack. Since the stacks are in parallel and we have 4 stacks, that gives a total current $i = 800 \cdot 4 = 3200\,\mathrm{A} = 3.2\,\mathrm{kA}$. In other words, option D.

▲

1.4 Electrolyzer dynamics

Figure 6 is taken from the group project, and depicts the anode water/O_2 gas separator system, see also Fig. 2.

The generation rates $\dot{n}^\mathrm{g}_j$ are given by Eqs. 23 and 24. Electro-osmotic water drag through the membrane of *a single cell* is given by

$$\dot{n}^\mathrm{m,d}_{\mathrm{H_2O}} = \eta_\mathrm{d} \cdot \frac{A_\mathrm{c} i''}{F}; \tag{30}$$

assume that η_d is known. Assume that $\dot{n}^\mathrm{i}_{\mathrm{H_2O}}$ is given (in the project: a P-controller was used) and $\dot{n}^\mathrm{c,r}_{\mathrm{H_2O}}$ is given (in the project: a P-controller was used). With water saturation pressure $p^\mathrm{sat}_{\mathrm{H_2O}}$ a known function of temperature T, the volumetric flow out of the top valve is given by

$$\dot{V}^\mathrm{e}_\mathrm{a} = \dot{V}^\varsigma_\mathrm{a} \frac{p_\mathrm{a} - p^\mathrm{e}_\mathrm{a}}{p^\circ} \tag{31}$$

where the total anode pressure p_a is

$$p_\mathrm{a} = p^\mathrm{a}_{\mathrm{O}_2} + p^\mathrm{sat}_{\mathrm{H_2O}}. \tag{32}$$

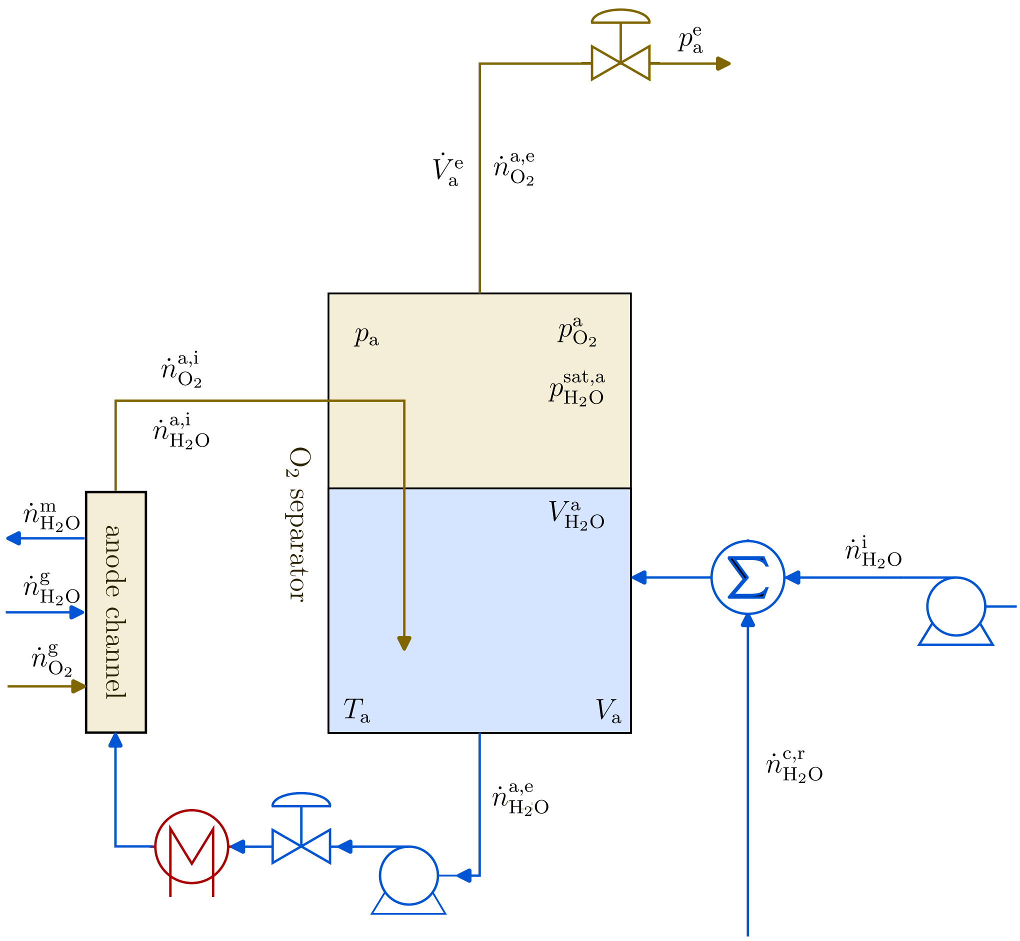

Figure 6: Anode oxygen-water separator subsystem. Note that generation $\dot{n}_{H_2O}^{g}$ is negative (consumption), which means that the actual flow is against the arrow direction.

Flows of oxygen and water vapor out with $\dot{V}_{\mathrm{a}}^{\mathrm{e}}$ are given by the flow-version of the ideal gas law

$$pV = nRT, \tag{33}$$

i.e., by

$$p_j\dot{V} = \dot{n}_jRT. \tag{34}$$

In the group project, water vapor leaving the anode tank via the gas valve (Eqs. 34, 31), $\dot{n}_{\mathrm{H_2O,gas}}^{\mathrm{a,e}}$, was neglected.

Problem 7. Anode tank water balance.

Using the symbols in Fig. 6 and the given expressions, formulate the mass balance for water in the separator tank, and find a differential equation for the water volume in the anode, $V_{\mathrm{H_2O}}^{\mathrm{a}}$, when assuming $\dot{n}_{\mathrm{H_2O,gas}}^{\mathrm{a,e}} = 0$.

	Problem 7: The water volume in the anode, $V_{\mathrm{H_2O}}^{\mathrm{a}}$, is given by the following differential equation — which answer is **correct**?	Choice
A:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} - \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} + N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	
B:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} - \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	
C:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	
D:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} + N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	

▲

Solution 7. Anode tank water balance.

Using the symbols in Fig. 6 and the given expressions, formulate the mass balance for water in the separator tank, and find a differential equation for the water volume in the anode, $V_{\mathrm{H_2O}}^{\mathrm{a}}$, when assuming $\dot{n}_{\mathrm{H_2O,gas}}^{\mathrm{a,e}} = 0$.

We have from Fig. 6:

$$\frac{\mathrm{d}n_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\dot{n}_{\mathrm{H_2O}}^{\mathrm{m,d}} + N_{\mathrm{c}}\dot{n}_{\mathrm{H_2O}}^{\mathrm{g}}$$

$$\Downarrow$$

$$\frac{\mathrm{d}n_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\eta_{\mathrm{d}}\cdot\frac{A_{\mathrm{c}}i''}{F} + N_{\mathrm{c}}\frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\frac{A_{\mathrm{c}}}{F}i''.$$

With $n_j = m_j/M_j$ and $m_j = \rho_jV$, we get

$$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \frac{\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}}{\rho_{\mathrm{H_2O}}/M_{\mathrm{H_2O}}}.$$

This means that *Choice C* is correct.

	Problem 7: The water volume in the anode, $V_{\mathrm{H_2O}}^{\mathrm{a}}$, is given by the following differential equation — which answer is **correct**?	Choice
A:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} - \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} + N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	
B:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} - \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	
C:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} - N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	×
D:	$\frac{\mathrm{d}V_{\mathrm{H_2O}}^{\mathrm{a}}}{\mathrm{d}t} = \left(\dot{n}_{\mathrm{H_2O}}^{\mathrm{i}} + \dot{n}_{\mathrm{H_2O}}^{\mathrm{c,r}} + N_{\mathrm{c}}\left(\eta_{\mathrm{d}} - \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}}\right)\cdot\frac{A_{\mathrm{c}}i''}{F}\right) / \frac{\rho_{\mathrm{H_2O}}}{M_{\mathrm{H_2O}}}$	

▲

Problem 8. Anode tank oxygen balance.

Using the symbols in Fig. 6 and additional expressions, formulate the mass balance for oxygen in the separator tank and find a model for the oxygen partial pressure $p^{\mathrm{a}}_{\mathrm{O_2}}$ in the anode tank.

	Problem 8: The oxygen pressure in the anode separator tank, $p^{\mathrm{a}}_{\mathrm{O_2}}$, is given by the following differential equation — which answer is **correct**? [Hint: $\frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t}$ is given in Problem 7].	Choice
A:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \left(p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{RT} \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right)$	
B:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \left(p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{RT} \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / V_{\mathrm{a}}$	
C:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \left(-p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{RT} \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right)$	
D:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \left(-p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{RT} \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / V_{\mathrm{a}}$	

▲

Solution 8. Anode tank oxygen balance.

Using the symbols in Fig. 6 and additional expressions, formulate the mass balance for oxygen in the separator tank and find a model for the oxygen partial pressure $p^{\mathrm{a}}_{\mathrm{O_2}}$ in the anode tank.

We have

$$\frac{\mathrm{d}n^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \dot{n}^{\mathrm{g}}_{\mathrm{O_2}} - \dot{n}^{\mathrm{a,e}}_{\mathrm{O_2}}$$

$$\Downarrow$$

$$\frac{\mathrm{d}n^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}} \dot{V}^{\mathrm{e}}_{\mathrm{a}}}{RT}$$

Here,

$$p_{\mathrm{O_2}} V^{\mathrm{a}}_{\mathrm{O_2}} = n^{\mathrm{a}}_{\mathrm{O_2}} RT$$

$$\Downarrow$$

$$n^{\mathrm{a}}_{\mathrm{O_2}} = \frac{p^{\mathrm{a}}_{\mathrm{O_2}} \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right)}{RT}$$

and

$$\dot{V}^{\mathrm{a}}_{\mathrm{a}} = \dot{V}^{\mathrm{e}}_{\mathrm{a}} = \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p_{\mathrm{a}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}} = \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}.$$

It follows that

$$\frac{\mathrm{d}n^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{p^{\mathrm{a}}_{\mathrm{O_2}} \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right)}{RT}\right) = \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right) \frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} - p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t}.$$

Thus

$$\left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right) \frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} - p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} = N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{RT} \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}$$

$$\Downarrow$$

$$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O_2}}}{\mathrm{d}t} = \frac{p^{\mathrm{a}}_{\mathrm{O_2}} \frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{c}}}{F} i'' - \frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{RT} \dot{V}^{\varsigma}_{\mathrm{a}} \frac{p^{\mathrm{a}}_{\mathrm{O_2}} + p^{\mathrm{sat}}_{\mathrm{H_2O}} - p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}}{V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}}.$$

In other words, *Choice A* is correct.

	Problem 8: The oxygen pressure in the anode separator tank, $p^{\mathrm{a}}_{\mathrm{O}_2}$, is given by the following differential equation — which answer is **correct**? [Hint: $\frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t}$ is given in Problem 7].	Choice
A:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O}_2}}{\mathrm{d}t} = \left(p^{\mathrm{a}}_{\mathrm{O}_2}\frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}}\frac{\nu_{\mathrm{O}_2}}{\nu_{\mathrm{e}^-}}\frac{A_{\mathrm{c}}}{F}i'' - \frac{p^{\mathrm{a}}_{\mathrm{O}_2}}{RT}\dot{V}^{\varsigma}_{\mathrm{a}}\frac{p^{\mathrm{a}}_{\mathrm{O}_2}+p^{\mathrm{sat}}_{\mathrm{H_2O}}-p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right)$	×
B:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O}_2}}{\mathrm{d}t} = \left(p^{\mathrm{a}}_{\mathrm{O}_2}\frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}}\frac{\nu_{\mathrm{O}_2}}{\nu_{\mathrm{e}^-}}\frac{A_{\mathrm{c}}}{F}i'' - \frac{p^{\mathrm{a}}_{\mathrm{O}_2}}{RT}\dot{V}^{\varsigma}_{\mathrm{a}}\frac{p^{\mathrm{a}}_{\mathrm{O}_2}+p^{\mathrm{sat}}_{\mathrm{H_2O}}-p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / V_{\mathrm{a}}$	
C:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O}_2}}{\mathrm{d}t} = \left(-p^{\mathrm{a}}_{\mathrm{O}_2}\frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}}\frac{\nu_{\mathrm{O}_2}}{\nu_{\mathrm{e}^-}}\frac{A_{\mathrm{c}}}{F}i'' - \frac{p^{\mathrm{a}}_{\mathrm{O}_2}}{RT}\dot{V}^{\varsigma}_{\mathrm{a}}\frac{p^{\mathrm{a}}_{\mathrm{O}_2}+p^{\mathrm{sat}}_{\mathrm{H_2O}}-p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / \left(V_{\mathrm{a}} - V^{\mathrm{a}}_{\mathrm{H_2O}}\right)$	
D:	$\frac{\mathrm{d}p^{\mathrm{a}}_{\mathrm{O}_2}}{\mathrm{d}t} = \left(-p^{\mathrm{a}}_{\mathrm{O}_2}\frac{\mathrm{d}V^{\mathrm{a}}_{\mathrm{H_2O}}}{\mathrm{d}t} + N_{\mathrm{c}}\frac{\nu_{\mathrm{O}_2}}{\nu_{\mathrm{e}^-}}\frac{A_{\mathrm{c}}}{F}i'' - \frac{p^{\mathrm{a}}_{\mathrm{O}_2}}{RT}\dot{V}^{\varsigma}_{\mathrm{a}}\frac{p^{\mathrm{a}}_{\mathrm{O}_2}+p^{\mathrm{sat}}_{\mathrm{H_2O}}-p^{\mathrm{e}}_{\mathrm{a}}}{p^{\circ}}\right) / V_{\mathrm{a}}$	

▲ Let $\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O,gas}}$ be the rate at which water is lost to the atmosphere through the anode effluent gas valve. In the group project, this loss of water was neglected.

The fractional water loss $x^{\mathrm{a}}_{\mathrm{H_2O}}$ loss can be defined as

$$x^{\mathrm{a}}_{\mathrm{H_2O}} \triangleq \frac{\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O,gas}}}{\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O,gas}} + \dot{n}^{\mathrm{a,e}}_{\mathrm{O}_2}}, \tag{35}$$

where the flow version of the ideal gas law is as in Eq. 34.

Problem 9. Anode tank pressure vs. water loss.

It is of interest to find whether, and if so, how the anode pressure p_{a} influences the fraction $x^{\mathrm{a}}_{\mathrm{H_2O}}$ in Eq. 35.

	Problem 9: The mole fraction of water transported through the anode gas exit valve is — which answer is **correct**?	Choice
A:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}}-p^{\mathrm{sat}}_{\mathrm{H_2O}}}$	
B:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p^{\mathrm{e}}_{\mathrm{a}}} = \text{constant}$	
C:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}}-p^{\mathrm{e}}_{\mathrm{a}}}$	
D:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}}}$	

▲

Solution 9. Anode tank pressure vs. water loss.

It is of interest to find whether the anode pressure p_{a} influences the fraction $x^{\mathrm{a}}_{\mathrm{H_2O}}$ in Eq. 35.

We find

$$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{\dot{n}^{\mathrm{gas}}_{\mathrm{H_2O}}}{\dot{n}^{\mathrm{gas}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{a,e}}_{\mathrm{O}_2}} = \frac{\frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}\dot{V}^{\mathrm{e}}_{\mathrm{a}}}{RT}}{\frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}\dot{V}^{\mathrm{e}}_{\mathrm{a}}}{RT} + \frac{p^{\mathrm{a}}_{\mathrm{O}_2}\dot{V}^{\mathrm{e}}_{\mathrm{a}}}{RT}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p^{\mathrm{sat}}_{\mathrm{H_2O}} + p_{\mathrm{a}} - p^{\mathrm{sat}}_{\mathrm{H_2O}}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}}}.$$

It follows that by increasing p_{a}, the fraction of water in the gas is reduced. This is an advantage. On the other hand, increased anode pressure requires extra energy usage. In other words, *Choice D* is correct.

	Problem 9: The mole fraction of water transported through the anode gas exit valve is — which answer is **correct**?	Choice
A:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}} - p^{\mathrm{sat}}_{\mathrm{H_2O}}}$	
B:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p^{\mathrm{e}}_{\mathrm{a}}} = \mathrm{constant}$	
C:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}} - p^{\mathrm{e}}_{\mathrm{a}}}$	
D:	$x^{\mathrm{a}}_{\mathrm{H_2O}} = \frac{p^{\mathrm{sat}}_{\mathrm{H_2O}}}{p_{\mathrm{a}}}$	×

▲

Problem 10. Anode tank time constant.

The transfer functions from current density i'' to anode tank oxygen pressure $p^{\mathrm{a}}_{\mathrm{O_2}}$ and water volume $\dot{V}^{\mathrm{a}}_{\mathrm{H_2O}}$ were given in the group project solution report as

$$\frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{i''} \propto \frac{(1+2.26s)\,(1+4.7s)}{(1+0.04s)\,(1+2.3s)\,(1+4.6s)} \tag{36}$$

$$\frac{V^{\mathrm{a}}_{\mathrm{H_2O}}}{i''} \propto \frac{1+23s}{(1+2.3s)\,(1+4.6s)} \tag{37}$$

with time given in minutes, and symbol "$\propto$" means "is proportional to" [i.e., $\frac{V^{\mathrm{a}}_{\mathrm{H_2O}}}{i''}$ is proportional to $\frac{1+23s}{(1+2.3s)(1+4.6s)}$, etc.].

	Problem 10: The time constants of the water/O_2 separator tank are — which answer is **correct?**	Choice
A:	$T_1 = 2.26\,\mathrm{min}$, $T_2 = 4.7\,\mathrm{min}$, $T_3 = 23\,\mathrm{min}$	
B:	$T_1 = 2.26\,\mathrm{min}$, $T_2 = 4.7\,\mathrm{min}$, $T_3 = 23\,\mathrm{min}$, $T_4 = 2.4\,\mathrm{s}$, $T_5 = 2.3\,\mathrm{min}$, $T_6 = 4.6\,\mathrm{min}$	
C:	$T_1 = 0.04\,\mathrm{min}$, $T_2 = 4.6\,\mathrm{min}$	
D:	$T_1 = 2.4\,\mathrm{s}$, $T_2 = 2.3\,\mathrm{min}$, $T_3 = 4.6\,\mathrm{min}$	

▲

Solution 10. Anode tank time constant.

The transfer functions from current density i'' to anode tank oxygen pressure $p^{\mathrm{a}}_{\mathrm{O_2}}$ and water volume $\dot{V}^{\mathrm{a}}_{\mathrm{H_2O}}$ were given in the group project solution report as

$$\frac{p^{\mathrm{a}}_{\mathrm{O_2}}}{i''} \propto \frac{(1+2.26s)\,(1+4.7s)}{(1+0.04s)\,(1+2.3s)\,(1+4.6s)}$$

$$\frac{V^{\mathrm{a}}_{\mathrm{H_2O}}}{i''} \propto \frac{1+23s}{(1+2.3s)\,(1+4.6s)}.$$

The time constants are given as $1/|s_{\mathrm{p}}|$ where s_{p} is a (real part) of the roots of the *denominator* of the transfer functions, i.e. by T in the factors $1+Ts$ of the *denominators* (s_{p} are the poles of the system). In other words, the time constants are $T_1 = 2.4\,\mathrm{s}, T_2 = 2.3\,\mathrm{min}, T_3 = 4.6\,\mathrm{min}$ (Choice D; $T_1 = 0.04\,\mathrm{min} = 0.04 \cdot 60\,\mathrm{s} = 2.4\,\mathrm{s}$). The roots s_0 of the numerator of the transfer functions are known as *zeros*, and are not time constants. Thus the zeros, given as $1/|s_0|$, are $T_1 = 2.26\,\mathrm{min}$, $T_2 = 4.7\,\mathrm{min}$, $T_3 = 23\,\mathrm{min}$ (Choice A), and these are not time constants.

Conceptually, one could approximately cancel the terms $(1+2.26s)\,/\,(1+2.3s)$ and $(1+4.7s)\,/\,(1+4.6s)$ and thus be left with only time constant $T = 0.04\,\mathrm{min} = 2.4\,\mathrm{s}$ in the transfer function from i'' to $p^{\mathrm{a}}_{\mathrm{O}_2}$. But the two other time constants show up in the other transfer function.

In summary, the correct answer is *Choice D*.

	Problem 10: The time constants of the water level in the tank are — which answer is **correct?**	Choice
A:	$T_1 = 2.26\,\mathrm{min}$, $T_2 = 4.7\,\mathrm{min}$, $T_3 = 23\,\mathrm{min}$	
B:	$T_1 = 2.26\,\mathrm{min}$, $T_2 = 4.7\,\mathrm{min}$, $T_3 = 23\,\mathrm{min}$, $T_4 = 2.4\,\mathrm{s}$, $T_5 = 2.3\,\mathrm{min}$, $T_6 = 4.6\,\mathrm{min}$	
C:	$T_1 = 0.04\,\mathrm{min}$, $T_2 = 4.6\,\mathrm{min}$	
D:	$T_1 = 2.4\,\mathrm{s}$, $T_2 = 2.3\,\mathrm{min}$, $T_3 = 4.6\,\mathrm{min}$	×

▲

1.5 Stack energy balance

In the group project, it was assumed that the temperature was constant ($T = 60\,^\circ\mathrm{C}$) and the same everywhere in the system. In reality, the temperature will vary, depending on how the system is operated.

Here, we will assume that the *stack* has homogeneous, possibly time varying temperature T_{s}, but that the rest of the system, including separator tanks, has constant temperature T. To find how T_{s} varies, we consider the energy balance for the stack.

The thermal energy balance can be written as

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \dot{H}_{\mathrm{i}} - \dot{H}_{\mathrm{e}} + \dot{W} + \dot{Q}, \tag{38}$$

where internal energy U is related to enthalpy H and pressure-volume pV as

$$U = H - pV. \tag{39}$$

Assuming an *ideal solution*, we have

$$H \approx \sum_j H_{\mathrm{j}} = \sum_j n_j \tilde{H}_j = \sum_j m_j \hat{H}_j \tag{40}$$

$$\dot{H} \approx \sum_j \dot{H}_j = \sum_j \dot{n}_j \tilde{H}_j = \sum_j \dot{m}_j \hat{H}_j; \tag{41}$$

in the expression for H, a term $n_j \tilde{H}_j$ may be replaced by $m_j \hat{H}_j$, and in the expression for $\dot{H}$, a term $\dot{n}_j \tilde{H}_j$ may be replaced by $\dot{m}_j \hat{H}_j$ if this is more convenient. The stack system consists of metal with mass m_{s} and heat capacity $\hat{c}_{\mathrm{p,s}}$, as well as fluids (water, hydrogen, oxygen) with amounts n_j and heat capacity $\tilde{c}_{\mathrm{p},j}$. For simplicity, assume that

$$\tilde{H}_j \approx \tilde{H}^\circ_j + \tilde{c}_{\mathrm{p},j}\left(T - T^\circ\right) \tag{42}$$

$$\hat{H}_j \approx \hat{H}^\circ_j + \hat{c}_{\mathrm{p},j}\left(T - T^\circ\right). \tag{43}$$

The main work rate $\dot{W}$ added to the system consists of electric power; possible friction loss will be neglected here:

$$\dot{W} = u_{\mathrm{s}} i_{\mathrm{s}} = N_{\mathrm{c}} u_{\mathrm{c}}\left(i'', T\right) i_{\mathrm{c}} \tag{44}$$

where i_c and i'' are related as in Eq. 19, and $u_c(i'', T)$ is the applied cell voltage u_c, Eq. 20. Added heat consists of convective heat $\dot{Q}_v$ and heat by radiation $\dot{Q}_r$,

$$\dot{Q} = \dot{Q}_v + \dot{Q}_r, \tag{45}$$

where

$$\dot{Q}_v = A_s \mathcal{U}_s (T_a - T_s) \tag{46}$$
$$\dot{Q}_r = A_s \varepsilon_s \sigma \left(T_a^4 - T_s^4\right); \tag{47}$$

A_s is the surface area of the stack, $\mathcal{U}_s$ is the stack overall heat transfer coefficient, T_a is the ambient temperature, T_s is the stack temperature, ε_s is the surface emmisivity, and σ is the Stefan-Boltzmann constant.

Problem 11. Stack energy balance — net enthalpy flow

In the energy balance, it is of interest to find an expression for the net enthalpy influent $\Delta\dot{H}$,

$$\Delta\dot{H} \triangleq \dot{H}_i - \dot{H}_e. \tag{48}$$

As part of $\Delta\dot{H}$, the enthalpy of reaction $\Delta_r\tilde{H}$ will appear,

$$\Delta_r\tilde{H} \triangleq \sum_j \nu_j \tilde{H}_j. \tag{49}$$

Show that the net enthalpy flow $\Delta\dot{H}$ through the stack can be expressed as

	Problem 11: Net enthalpy flow $\Delta\dot{H}$ through the stack can be expressed as — which expression is **correct**?	Choice
A:	$\Delta\dot{H} = -N_c \frac{\Delta_r\tilde{H}}{F\nu_{e^-}} A_c i''$	
B:	$\Delta\dot{H} = \dot{n}^{a,e}_{H_2O}\tilde{c}_{p,H_2O}(T - T_s) - N_c \frac{\Delta_r\tilde{H}}{F\nu_{e^-}} A_c i''$	
C:	$\Delta\dot{H} = \dot{n}^{a,e}_{H_2O}\tilde{c}_{p,H_2O}(T - T_s) + N_c \frac{\Delta_r\tilde{H}}{F\nu_{e^-}} A_c i''$	
D:	$\Delta\dot{H} = -\dot{n}^{a,e}_{H_2O}\tilde{c}_{p,H_2O}(T - T_s) - N_c \frac{\Delta_r\tilde{H}}{F\nu_{e^-}} A_c i''$	

▲

Solution 11. Stack energy balance — net enthalpy flow

In the energy balance, it is of interest to find an expression for the net enthalpy influent $\Delta\dot{H}$,

$$\Delta\dot{H} \triangleq \dot{H}_i - \dot{H}_e.$$

As part of $\Delta\dot{H}$, the enthalpy of reaction $\Delta_r\tilde{H}$ will appear,

$$\Delta_r\tilde{H} \triangleq \sum_j \nu_j \tilde{H}_j.$$

On the RHS[4] of the thermal energy balance, we have the net enthalpy flow $\Delta\dot{H}$,

$$\Delta\dot{H} \triangleq \dot{H}_i - \dot{H}_e.$$

[4]RHS = right hand side

We have

$$\dot{H}_{\mathrm{i}} = \dot{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} = \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}}$$

and

$$\begin{aligned}
\dot{H}_{\mathrm{e}} &= \dot{H}^{\mathrm{a,i}}_{\mathrm{H_2O}} + \dot{H}^{\mathrm{c,i}}_{\mathrm{H_2O}} + \dot{H}^{\mathrm{a,i}}_{\mathrm{O_2}} + \dot{H}^{\mathrm{c,i}}_{\mathrm{H_2}} \\
&\Downarrow \\
\dot{H}_{\mathrm{e}} &= \dot{n}^{\mathrm{a,i}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{c,i}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{a,i}}_{\mathrm{O_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{O_2}} + \dot{n}^{\mathrm{c,i}}_{\mathrm{H_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2}}
\end{aligned}$$

where superscript s in $\tilde{H}^{\mathrm{s}}_j$ refers to the stack temperature T. The net enthalpy flow is then

$$\Delta \dot{H} = \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \left(\dot{n}^{\mathrm{a,i}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{c,i}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{a,i}}_{\mathrm{O_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{O_2}} + \dot{n}^{\mathrm{c,i}}_{\mathrm{H_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2}} \right).$$

Furthermore, from steady mass balances, we have

$$\begin{aligned}
\dot{n}^{\mathrm{a,i}}_{\mathrm{H_2O}} &= \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{g}}_{\mathrm{H_2O}} - \dot{n}^{\mathrm{m}}_{\mathrm{H_2O}} \\
\dot{n}^{\mathrm{c,i}}_{\mathrm{H_2O}} &= \dot{n}^{\mathrm{m}}_{\mathrm{H_2O}} \\
\dot{n}^{\mathrm{a,i}}_{\mathrm{O_2}} &= \dot{n}^{\mathrm{g}}_{\mathrm{O_2}} \\
\dot{n}^{\mathrm{c,i}}_{\mathrm{H_2}} &= \dot{n}^{\mathrm{g}}_{\mathrm{H_2}}.
\end{aligned}$$

This gives:

$$\begin{aligned}
\Delta \dot{H} &= \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \left(\left(\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{g}}_{\mathrm{H_2O}} - \dot{n}^{\mathrm{m}}_{\mathrm{H_2O}} \right) \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{m}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{g}}_{\mathrm{O_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{O_2}} + \dot{n}^{\mathrm{g}}_{\mathrm{H_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2}} \right) \\
&\Downarrow \\
\Delta \dot{H} &= \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \left(\tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} \right) - \left(\dot{n}^{\mathrm{g}}_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \dot{n}^{\mathrm{g}}_{\mathrm{O_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{O_2}} + \dot{n}^{\mathrm{g}}_{\mathrm{H_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2}} \right).
\end{aligned}$$

Here, we have

$$\begin{aligned}
\dot{n}^{\mathrm{g}}_{\mathrm{H_2O}} &= N_{\mathrm{c}} \frac{\nu_{\mathrm{H_2O}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{m}} i''}{F} \\
\dot{n}^{\mathrm{g}}_{\mathrm{H_2}} &= N_{\mathrm{c}} \frac{\nu_{\mathrm{H_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{m}} i''}{F} \\
\dot{n}^{\mathrm{g}}_{\mathrm{O_2}} &= N_{\mathrm{c}} \frac{\nu_{\mathrm{O_2}}}{\nu_{\mathrm{e^-}}} \frac{A_{\mathrm{m}} i''}{F},
\end{aligned}$$

and we can simplify the expression to

$$\Delta \dot{H} = \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \left(\tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} \right) - N_{\mathrm{c}} \frac{A_{\mathrm{m}} i''}{\nu_{\mathrm{e^-}} F} \left(\nu_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \nu_{\mathrm{O_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{O_2}} + \nu_{\mathrm{H_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2}} \right).$$

Here, we can introduce the enthalpy of reaction,

$$\Delta_{\mathrm{r}} \tilde{H} \triangleq \nu_{\mathrm{H_2O}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} + \nu_{\mathrm{O_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{O_2}} + \nu_{\mathrm{H_2}} \tilde{H}^{\mathrm{s}}_{\mathrm{H_2}} = \sum_j \nu_j \tilde{H}^{\mathrm{s}}_j.$$

With the expression for liquid molar enthalpy, we find

$$\begin{aligned}
\tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} &= \tilde{H}^{\circ}_{\mathrm{H_2O}} + \tilde{c}_{\mathrm{p,H_2O}} \left(T - T^{\circ} \right) + \tilde{V}_{\mathrm{H_2O}} \left(p_{\mathrm{a,e}} - p^{\circ} \right) \\
&\quad - \tilde{H}^{\circ}_{\mathrm{H_2O}} - \tilde{c}_{\mathrm{p,H_2O}} \left(T_{\mathrm{s}} - T^{\circ} \right) - \tilde{V}_{\mathrm{H_2O}} \left(p_{\mathrm{a,s}} - p^{\circ} \right) \\
&\Downarrow \\
\tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} &= \tilde{c}_{\mathrm{p,H_2O}} \left(T - T_{\mathrm{s}} \right) + \tilde{V}_{\mathrm{H_2O}} \left(p_{\mathrm{a,e}} - p_{\mathrm{a,s}} \right).
\end{aligned}$$

Here, $p_{\mathrm{a,s}}$ is the pressure in the stack at the anode side. Although there is some friction pressure drop into the anode channel, we will assume that $p_{\mathrm{a,s}} \approx p_{\mathrm{a,e}}$, and the expression simplifies to

$$\tilde{H}^{\mathrm{a,e}}_{\mathrm{H_2O}} - \tilde{H}^{\mathrm{s}}_{\mathrm{H_2O}} = \tilde{c}_{\mathrm{p,H_2O}} \left(T - T_{\mathrm{s}}\right).$$

We can thus write $\Delta\dot{H}$ as

$$\Delta\dot{H} = \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{c}_{\mathrm{p,H_2O}} \left(T - T_{\mathrm{s}}\right) - N_{\mathrm{c}} \frac{\Delta_{\mathrm{r}}\tilde{H}}{F\nu_{\mathrm{e^-}}} A_{\mathrm{m}} i''.$$

In other words, *Choice B* is correct.

	Problem 11: Net enthalpy flow $\Delta\dot{H}$ through the stack can be expressed as — which expression is **correct**?	Choice
A:	$\Delta\dot{H} = -N_{\mathrm{c}} \frac{\Delta_{\mathrm{r}}\tilde{H}}{F\nu_{\mathrm{e^-}}} A_{\mathrm{c}} i''$	
B:	$\Delta\dot{H} = \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{c}_{\mathrm{p,H_2O}} \left(T - T_{\mathrm{s}}\right) - N_{\mathrm{c}} \frac{\Delta_{\mathrm{r}}\tilde{H}}{F\nu_{\mathrm{e^-}}} A_{\mathrm{c}} i''$	×
C:	$\Delta\dot{H} = \dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{c}_{\mathrm{p,H_2O}} \left(T - T_{\mathrm{s}}\right) + N_{\mathrm{c}} \frac{\Delta_{\mathrm{r}}\tilde{H}}{F\nu_{\mathrm{e^-}}} A_{\mathrm{c}} i''$	
D:	$\Delta\dot{H} = -\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}} \tilde{c}_{\mathrm{p,H_2O}} \left(T - T_{\mathrm{s}}\right) - N_{\mathrm{c}} \frac{\Delta_{\mathrm{r}}\tilde{H}}{F\nu_{\mathrm{e^-}}} A_{\mathrm{c}} i''$	

▲

Similarly to the Nernst equilibrium voltage u_{n} which can be expressed as in Eq. 10, $u_{\mathrm{n}} = \frac{\Delta_{\mathrm{r}}\tilde{G}}{F\nu_{\mathrm{e^-}}}$, we can define the so-called *thermoneutral* voltage u_{th} as

$$u_{\mathrm{th}} \triangleq \frac{\Delta_{\mathrm{r}}\tilde{H}}{F\nu_{\mathrm{e^-}}}. \tag{50}$$

Observe that

$$\Delta_{\mathrm{r}}\tilde{H} = \Delta_{\mathrm{r}}\tilde{G} + T\Delta_{\mathrm{r}}\tilde{S} > \Delta_{\mathrm{r}}\tilde{G} \Rightarrow u_{\mathrm{th}} > u_{\mathrm{n}}. \tag{51}$$

Problem 12. Stack energy balance — ODE

Assume that the internal energy U is approximately equal to the internal energy of metal, and assume that pV = constant. Show that the differential equation for the temperature can be written as

	Problem 12: Ordinary differential equation for stack temperature T_{s} — which expression is **correct**?	Choice
A:	$\frac{\mathrm{d}T_{\mathrm{s}}}{\mathrm{d}t} = \frac{\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_{\mathrm{s}})+N_{\mathrm{c}}A_{\mathrm{c}}i''(u_{\mathrm{th}}-u_{\mathrm{c}})+A_{\mathrm{s}}\left(\mathcal{U}_{\mathrm{s}}(T_{\mathrm{a}}-T_{\mathrm{s}})+\varepsilon_{\mathrm{s}}\sigma\left(T_{\mathrm{a}}^4-T_{\mathrm{s}}^4\right)\right)}{m_{\mathrm{s}}}$	
B:	$\frac{\mathrm{d}T_{\mathrm{s}}}{\mathrm{d}t} = \frac{\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_{\mathrm{s}})+N_{\mathrm{c}}A_{\mathrm{c}}i''(u_{\mathrm{c}}-u_{\mathrm{th}})+A_{\mathrm{s}}\left(\mathcal{U}_{\mathrm{s}}(T_{\mathrm{a}}-T_{\mathrm{s}})+\varepsilon_{\mathrm{s}}\sigma\left(T_{\mathrm{a}}^4-T_{\mathrm{s}}^4\right)\right)}{m_{\mathrm{s}}\hat{c}_{\mathrm{p,s}}}$	
C:	$\frac{\mathrm{d}T_{\mathrm{s}}}{\mathrm{d}t} = \frac{\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_{\mathrm{s}})+N_{\mathrm{c}}A_{\mathrm{c}}i''(u_{\mathrm{c}}-u_{\mathrm{th}})+A_{\mathrm{s}}\left(\mathcal{U}_{\mathrm{s}}(T_{\mathrm{a}}-T_{\mathrm{s}})+\varepsilon_{\mathrm{s}}\sigma\left(T_{\mathrm{a}}^4-T_{\mathrm{s}}^4\right)\right)}{m_{\mathrm{s}}}$	
D:	$\frac{\mathrm{d}T_{\mathrm{s}}}{\mathrm{d}t} = \frac{\dot{n}^{\mathrm{a,e}}_{\mathrm{H_2O}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_{\mathrm{s}})-N_{\mathrm{c}}A_{\mathrm{c}}i''(u_{\mathrm{c}}-u_{\mathrm{th}})+A_{\mathrm{s}}\left(\mathcal{U}_{\mathrm{s}}(T_{\mathrm{a}}-T_{\mathrm{s}})+\varepsilon_{\mathrm{s}}\sigma\left(T_{\mathrm{a}}^4-T_{\mathrm{s}}^4\right)\right)}{m_{\mathrm{s}}\hat{c}_{\mathrm{p,s}}}$	

▲

Solution 12. Stack energy balance — ODE

For U, we have

$$U = m_{\mathrm{s}} \left(\hat{H}^{\circ}_{\mathrm{s}} + \hat{c}_{\mathrm{p,s}} \left(T_{\mathrm{s}} - T^{\circ}\right)\right);$$

the values of m_s and $\hat{c}_\mathrm{p,s}$ must be provided; $\hat{H}_\mathrm{s}^\circ$ and T° will cancel out in the differentiation, and their accurate values are not important. Thus, at the LHS of the energy balance, we get

$$\frac{\mathrm{d}U}{\mathrm{d}t} \approx m_\mathrm{s}\hat{c}_\mathrm{p,s}\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t}.$$

By inserting the expression for $\Delta\dot{H}$ found in Problem 11, as well as the provides expressions for $\dot{Q}$ and $\dot{W} = N_\mathrm{c} = u_\mathrm{s}i_\mathrm{s} = N_\mathrm{c}u_\mathrm{c}i_\mathrm{c} = N_\mathrm{c}u_\mathrm{c}A_\mathrm{c}i''$, we find

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \Delta\dot{H} + \dot{W} + \dot{Q}$$

$$\Downarrow$$

$$m_\mathrm{s}\hat{c}_\mathrm{p,s}\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}\left(T - T_\mathrm{s}\right) - N_\mathrm{c}\frac{\Delta_\mathrm{r}\tilde{H}}{F\nu_{\mathrm{e}^-}}A_\mathrm{c}i'' + N_\mathrm{c}u_\mathrm{c}A_\mathrm{c}i'' + A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}\left(T_\mathrm{a} - T_\mathrm{s}\right) + \varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4 - T_\mathrm{s}^4\right)\right).$$

By introducing

$$u_\mathrm{th} = \frac{\Delta_\mathrm{r}\tilde{H}}{F\nu_{\mathrm{e}^-}}$$

we then get

$$m_\mathrm{s}\hat{c}_\mathrm{p,s}\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}\left(T - T_\mathrm{s}\right) - N_\mathrm{c}u_\mathrm{th}A_\mathrm{c}i'' + N_\mathrm{c}u_\mathrm{c}A_\mathrm{c}i'' + A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}\left(T_\mathrm{a} - T_\mathrm{s}\right) + \varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4 - T_\mathrm{s}^4\right)\right)$$

$$\Downarrow$$

$$m_\mathrm{s}\hat{c}_\mathrm{p,s}\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}\left(T - T_\mathrm{s}\right) + N_\mathrm{c}A_\mathrm{c}i''\left(u_\mathrm{c} - u_\mathrm{th}\right) + A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}\left(T_\mathrm{a} - T_\mathrm{s}\right) + \varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4 - T_\mathrm{s}^4\right)\right)$$

and we see that *Choice B* is correct.

	Problem 12: Ordinary differential equation for stack temperature T_s — which expression is **correct**?	Choice
A:	$\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \frac{\dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_\mathrm{s})+N_\mathrm{c}A_\mathrm{c}i''(u_\mathrm{th}-u_\mathrm{c})+A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}(T_\mathrm{a}-T_\mathrm{s})+\varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4-T_\mathrm{s}^4\right)\right)}{m_\mathrm{s}}$	
B:	$\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \frac{\dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_\mathrm{s})+N_\mathrm{c}A_\mathrm{c}i''(u_\mathrm{c}-u_\mathrm{th})+A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}(T_\mathrm{a}-T_\mathrm{s})+\varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4-T_\mathrm{s}^4\right)\right)}{m_\mathrm{s}\hat{c}_\mathrm{p,s}}$	×
C:	$\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \frac{\dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_\mathrm{s})+N_\mathrm{c}A_\mathrm{c}i''(u_\mathrm{c}-u_\mathrm{th})+A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}(T_\mathrm{a}-T_\mathrm{s})+\varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4-T_\mathrm{s}^4\right)\right)}{m_\mathrm{s}}$	
D:	$\frac{\mathrm{d}T_\mathrm{s}}{\mathrm{d}t} = \frac{\dot{n}_{\mathrm{H_2O}}^{\mathrm{a,e}}\tilde{c}_{\mathrm{p,H_2O}}(T-T_\mathrm{s})-N_\mathrm{c}A_\mathrm{c}i''(u_\mathrm{c}-u_\mathrm{th})+A_\mathrm{s}\left(\mathcal{U}_\mathrm{s}(T_\mathrm{a}-T_\mathrm{s})+\varepsilon_\mathrm{s}\sigma\left(T_\mathrm{a}^4-T_\mathrm{s}^4\right)\right)}{m_\mathrm{s}\hat{c}_\mathrm{p,s}}$	

▲

The cell voltage u_c depends on current density i'' and temperature T_s, while the thermoneutral voltage u_th only depends on temperature T_s. Figure 7 illustrates u_c and u_th as a function of current density i''.

Problem 13. The role of thermoneutral voltage

The value of the thermoneutral voltage u_th may or may not influence the stack temperature T_s. Which of the following statements is correct?

	Problem 13: How does the thermoneutral voltage affect the stack temperature — which statement is **correct**?	Choice
A:	The thermoneutral voltage u_th has no effect on the stack temperature.	
B:	The thermoneutral voltage u_th is a fictitious voltage that is only used in efficiency computations.	
C:	If the cell voltage u_c is smaller than the thermoneutral voltage u_th, $u_\mathrm{c} < u_\mathrm{th}$, this will heat the stack.	
D:	If the cell voltage u_c is larger than the thermoneutral voltage u_th, $u_\mathrm{c} > u_\mathrm{th}$, this will heat the stack.	

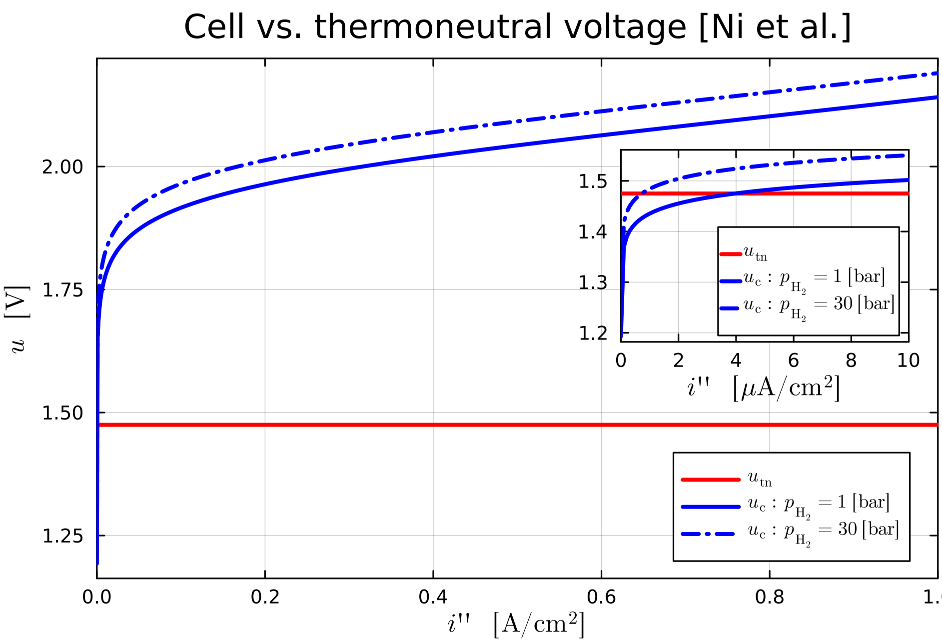

Figure 7: Comparing temperature variation in thermoneutral voltage u_{tn} and cell voltage u_{c}.

▲

Solution 13. The role of thermoneutral voltage

From the correct choice of the energy balance, we see that

$$m_{\text{s}}\hat{c}_{\text{p,s}}\frac{\mathrm{d}T_{\text{s}}}{\mathrm{d}t} = \dot{n}^{\text{a,e}}_{\text{H}_2\text{O}}\tilde{c}_{\text{p,H}_2\text{O}}\left(T - T_{\text{s}}\right) + N_{\text{c}}A_{\text{c}}i''\left(u_{\text{c}} - u_{\text{th}}\right) + A_{\text{s}}\left(\mathcal{U}_{\text{s}}\left(T_{\text{a}} - T_{\text{s}}\right) + \varepsilon_{\text{s}}\sigma\left(T_{\text{a}}^4 - T_{\text{s}}^4\right)\right).$$

In other words, if $u_{\text{c}} > u_{\text{th}}$, this will make $\frac{\mathrm{d}T_{\text{s}}}{\mathrm{d}t}$ more positive, hence cause T_{s} to increase, hence heat the stack. This means that *Choice D* is correct. All the other choices are incorrect.

	Problem 13: How does the thermoneutral voltage affect the stack temperature — which statement is **correct**?	Choice
A:	The thermoneutral voltage u_{th} has no effect on the stack temperature.	
B:	The thermoneutral voltage u_{th} is a fictitious voltage that is only used in efficiency computations.	
C:	If the cell voltage u_{c} is smaller than the thermoneutral voltage u_{th}, $u_{\text{c}} < u_{\text{th}}$, this will heat the stack.	
D:	If the cell voltage u_{c} is larger than the thermoneutral voltage u_{th}, $u_{\text{c}} > u_{\text{th}}$, this will heat the stack.	×

▲

Using the group project case as an example, but keeping the water in the anode tank at $T = T_{\text{sep}} = 50\,°\text{C}$ while *reducing* the current density i'' at $t = 10\,\text{min}$, *reducing* the anode channel circulation $\dot{n}^{\text{a,e}}_{\text{H}_2\text{O}}$ at $t = 20\,\text{min}$, and *increasing* the ambient temperature T_{a} at $t = 40\,\text{min}$ the temperature, the stack temperature T_{s} varies as in Fig. 8

In Fig. 8, the initial increase in T_{s} (at $t = 0$) is the result of starting the simulation away from steady state.

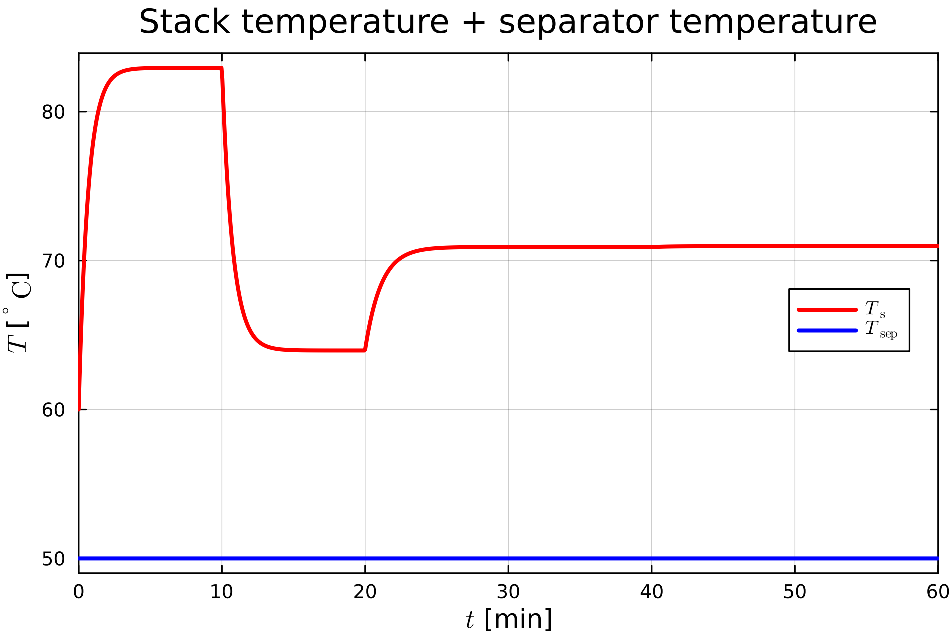

Figure 8: Stack temperature T_s and assumed perfectly constant separator volume temperatures $T_{sep} = T$.

Problem 14. Thermal time constant

Give an estimate of the thermal time constant of the system based on Fig. 8.

	Problem 14: The thermal time constant of the stack is — which statement is **correct**?	Choice
A:	It is not possible to assess the thermal time constant from Fig. 8.	
B:	The thermal time constant is approximately 2 min	
C:	The thermal time constant is approximately 10 min	
D:	The thermal time constant is approximately 25 min	

▲

Solution 14. Thermal time constant

Based on Fig. 8, the thermal time constant is approximately 2 min. This means that *Choice B* is correct.

	Problem 14: The thermal time constant of the stack is — which statement is **correct**?	Choice
A:	It is not possible to assess the thermal time constant from Fig. 8.	
B:	The thermal time constant is approximately 2 min	×
C:	The thermal time constant is approximately 10 min	
D:	The thermal time constant is approximately 25 min	

▲

1.6 Stack electrodynamics

In the group project, possible dynamics in the electric part of the membrane has been neglected. In reality, on the electrode interface between the catalyst and the membrane, there is a so-called *double layer* capacitor with capacitance C_{dl}, one for the anode (C_{dl}^{a})

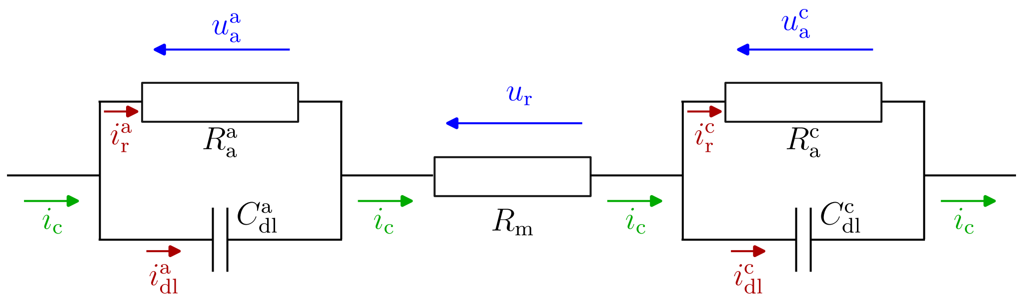

Figure 9: Equivalent linear electric model for kinetics of membrane.

and one for the cathode ($C_{\rm dl}^{\rm c}$). For simplicity, assume that the capacitance is the same for both electrodes, i.e., $C_{\rm dl}^{\rm a} = C_{\rm dl}^{\rm c} = C_{\rm dl}$. It is common to express $C_{\rm dl}$ as

$$C_{\rm dl} = A_{\rm c} C''_{\rm dl}, \tag{52}$$

where $C''_{\rm dl}$ is the double-layer capacitance per cell membrane area $A_{\rm c}$.

The capacitors are electrically in parallel with the activation voltage drops $u_{\rm a}$, with $u_{\rm a}^{\rm a}$ for the anode (OER, oxygen reaction), and $u_{\rm a}^{\rm c}$ for the cathode (HER, hydrogen reaction). The activation voltages $u_{\rm a}$ are functions of current density i'', and therefore of current $i_{\rm c}$.

From an electrical engineering point of view, this means that the relationship $u_{\rm a} = u_{\rm a}(i'')$ can be considered a non-linear resistor voltage drop, in other words: by linearizing this voltage drop around the operating point, we get $u_{\rm a}^{\delta} \approx R_{\rm a} i_{\rm r}^{\delta}$ where $u_{\rm a}^{\delta}$ and $i_{\rm r}^{\delta}$ are deviations from their respective operating points. Thus, one parallel RC circuit at both of the electrodes ($C_{\rm dl}^{\rm a} = C_{\rm dl}$, $R_{\rm a}^{\rm a}$, and $C_{\rm dl}^{\rm c} = C_{\rm dl}$, $R_{\rm a}^{\rm c}$), separated by the membrane resistor $R_{\rm m}$, Fig. 9.

Figure 9 shows the relevant currents $i_{\rm c}$ (green color), $i_{\rm r}^{\rm a}$, $i_{\rm dl}^{\rm a}$, $i_{\rm r}^{\rm c}$, and $i_{\rm dl}^{\rm c}$ (red color), as well as relevant voltages $u_{\rm a}^{\rm a}$, $u_{\rm r}$, and $u_{\rm a}^{\rm c}$ (blue color). Because the depicted circuit in Fig. 9 is linear, the order of the blocks is irrelevant, and we will neglect the membrane resistor $R_{\rm m}$ in the sequel; we will set the *total* activation voltage $u_{\rm a}$ equal to the *sum* of activation voltage losses on the anode ($u_{\rm a}^{\rm a}$) and the cathode ($u_{\rm a}^{\rm c}$).

Because the course syllabus for course FM1015 Modelling of Dynamic Systems does not include modeling of electric circuits, the model for the activation voltages is given below for convenience:

$$C_{\rm dl}^{\rm a} \frac{{\rm d}u_{\rm a}^{\rm a}}{{\rm d}t} = i_{\rm dl}^{\rm a} \tag{53}$$

$$u_{\rm a}^{\rm a} = R_{\rm a}^{\rm a} i_{\rm r}^{\rm a} \tag{54}$$

$$i_{\rm c} = i_{\rm r}^{\rm a} + i_{\rm dl}^{\rm a} \tag{55}$$

$$C_{\rm dl}^{\rm c} \frac{{\rm d}u_{\rm a}^{\rm c}}{{\rm d}t} = i_{\rm dl}^{\rm c} \tag{56}$$

$$u_{\rm a}^{\rm c} = R_{\rm a}^{\rm c} i_{\rm r}^{\rm a} \tag{57}$$

$$i_{\rm c} = i_{\rm r}^{\rm c} + i_{\rm dl}^{\rm c} \tag{58}$$

$$u_{\rm a} = u_{\rm a}^{\rm a} + u_{\rm a}^{\rm c}. \tag{59}$$

Problem 15. Electrodynamics — transfer function

It is of interest to find the *transfer function* $h(s)$ from external current $i_{\rm c}$ to total activation voltage loss $u_{\rm a}$, i.e.,

$$u_{\rm a}(s) = h(s) \cdot i_{\rm c}(s). \tag{60}$$

	Problem 15: Which expression for the transfer function from $i_{\mathrm{c}}(s)$ to $u_{\mathrm{a}}(s)$, denoted $h(s)$, is ***correct***?	Choice
A:	$h(s) = \frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s} + \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s}$	
B:	$h(s) = \frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s} \cdot \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s}$	
C:	$h(s) = \left(R_{\mathrm{a}}^{\mathrm{a}} + R_{\mathrm{a}}^{\mathrm{c}}\right) \frac{1+\left(C_{\mathrm{dl}}^{\mathrm{a}}+C_{\mathrm{dl}}^{\mathrm{c}}\right)s}{\left(1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s\right)\left(1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s\right)}$	
D:	$h(s) = \left(C_{\mathrm{dl}}^{\mathrm{a}} + C_{\mathrm{dl}}^{\mathrm{c}}\right) \frac{R_{\mathrm{a}}^{\mathrm{a}}R_{\mathrm{a}}^{\mathrm{c}}+\left(R_{\mathrm{a}}^{\mathrm{a}}+R_{\mathrm{a}}^{\mathrm{c}}\right)s}{\left(1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s\right)\left(1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s\right)}$	

Solution 15. Electrodynamics — transfer function

The anode and cathode circuits are identical in structure. Considering the anode circuit, Laplace transform of this gives

$$C_{\mathrm{dl}}^{\mathrm{a}}\frac{\mathrm{d}u_{\mathrm{a}}^{\mathrm{a}}}{\mathrm{d}t} = i_{\mathrm{dl}}^{\mathrm{a}} \Rightarrow sC_{\mathrm{dl}}^{\mathrm{a}}u_{\mathrm{a}}^{\mathrm{a}}(s) = i_{\mathrm{dl}}^{\mathrm{a}}(s)$$

and

$$u_{\mathrm{a}}^{\mathrm{a}} = R_{\mathrm{a}}^{\mathrm{a}}i_{\mathrm{r}}^{\mathrm{a}} \Rightarrow i_{\mathrm{r}}^{\mathrm{a}}(s) = \frac{1}{R_{\mathrm{a}}^{\mathrm{a}}}u_{\mathrm{a}}^{\mathrm{a}}(s).$$

Finally,

$$i_{\mathrm{c}}(s) = i_{\mathrm{r}}^{\mathrm{a}}(s) + i_{\mathrm{dl}}^{\mathrm{a}}(s) = \frac{1}{R_{\mathrm{a}}^{\mathrm{a}}}u_{\mathrm{a}}^{\mathrm{a}}(s) + sC_{\mathrm{dl}}^{\mathrm{a}}u_{\mathrm{a}}^{\mathrm{a}}(s) = \frac{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s}{R_{\mathrm{a}}^{\mathrm{a}}}u_{\mathrm{a}}^{\mathrm{a}}(s)$$
$$\Downarrow$$
$$u_{\mathrm{a}}^{\mathrm{a}}(s) = \frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s}i_{\mathrm{c}}(s).$$

Because of identical structure for the cathode circuit, we find

$$u_{\mathrm{a}}^{\mathrm{c}}(s) = \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s}i_{\mathrm{c}}(s).$$

Finally, we have

$$u_{\mathrm{a}} = u_{\mathrm{a}}^{\mathrm{a}} + u_{\mathrm{a}}^{\mathrm{c}}$$
$$\Downarrow$$
$$u_{\mathrm{a}}(s) = \left(\frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s} + \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s}\right)i_{\mathrm{c}}(s)$$
$$\Downarrow$$
$$\frac{u_{\mathrm{a}}(s)}{i_{\mathrm{c}}(s)} = \frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s} + \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s} = h(s).$$

It follows that *Choice A* is correct.

	Problem 15: Which expression for the transfer function from $i_{\mathrm{c}}(s)$ to $u_{\mathrm{a}}(s)$, denoted $h(s)$, is ***correct***?	Choice
A:	$h(s) = \frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s} + \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s}$	×
B:	$h(s) = \frac{R_{\mathrm{a}}^{\mathrm{a}}}{1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s} \cdot \frac{R_{\mathrm{a}}^{\mathrm{c}}}{1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s}$	
C:	$h(s) = \left(R_{\mathrm{a}}^{\mathrm{a}} + R_{\mathrm{a}}^{\mathrm{c}}\right) \frac{1+\left(C_{\mathrm{dl}}^{\mathrm{a}}+C_{\mathrm{dl}}^{\mathrm{c}}\right)s}{\left(1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s\right)\left(1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s\right)}$	
D:	$h(s) = \left(C_{\mathrm{dl}}^{\mathrm{a}} + C_{\mathrm{dl}}^{\mathrm{c}}\right) \frac{R_{\mathrm{a}}^{\mathrm{a}}R_{\mathrm{a}}^{\mathrm{c}}+\left(R_{\mathrm{a}}^{\mathrm{a}}+R_{\mathrm{a}}^{\mathrm{c}}\right)s}{\left(1+R_{\mathrm{a}}^{\mathrm{a}}C_{\mathrm{dl}}^{\mathrm{a}}s\right)\left(1+R_{\mathrm{a}}^{\mathrm{c}}C_{\mathrm{dl}}^{\mathrm{c}}s\right)}$	

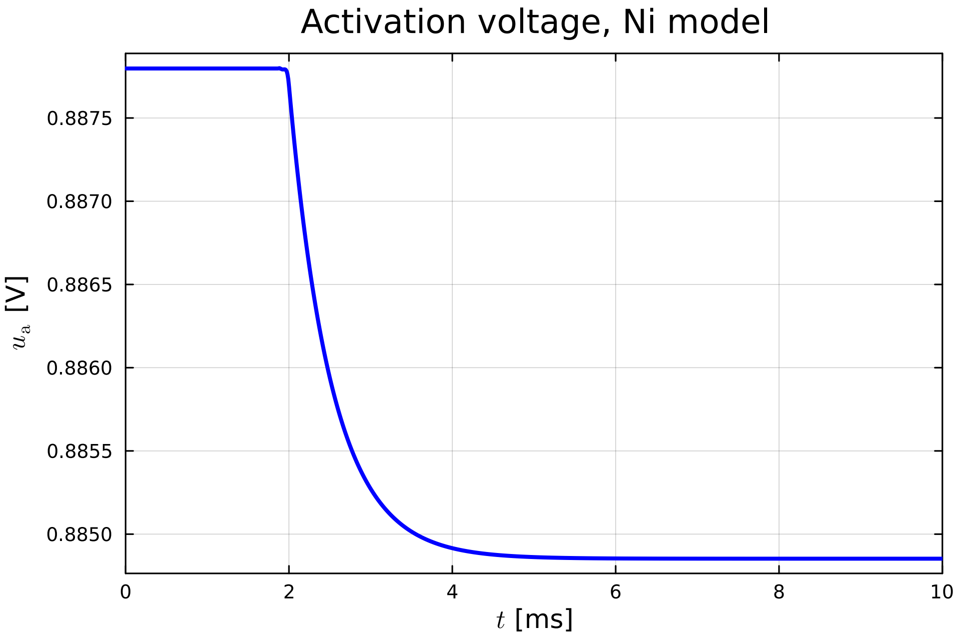

Figure 10: Activation voltage u_a after a step change in current density i''.

▲

Figure 10 shows the step-response in combined activation voltage $u_a = u_a^a + u_a^c$ after a step change in i''; note that $i'' \propto i_c$. NOTE! Different activation voltage models (Butler-Volmer) may give quite different (e.g., 10 times slower) time responses.

Problem 16. Electrodynamics — time constant

Based on Fig. 10, give an estimate of the time constant T_e of the electric (sub-) system.

	Problem 16: The time constant T_e of the electric subsystem is — which answer is ***correct***?	Choice
A:	The variation in u_a is too small to determine the time constant	
B:	$T_e = 4\,\text{ms}$	
C:	$T_e = 2\,\text{ms}$	
D:	$T_e = 0.5\,\text{ms}$	

▲

Solution 16. Electrodynamics — time constant

Based on Fig. 10, the time constant for the electric subsystem is approximately $T_e \approx 0.5\,\text{ms}$, which is quite fast.

It is interesting to observe that Fig 10 seems to indicate a single time constant, while one should expect two time constants $T_e^a = R_a^a C_{dl}^a$ and $T_e^c = R_a^c C_{dl}^c$. The explanation is quite simple: in the model, we have assumed $C_{dl} = C_{dl}^a + C_{dl}^c$. It turns out that because of the particular values in the Butler-Volmer model of the group project, $R_a^a \approx R_a^c = R_a$, hence $T_e^a = T_e^c = T_e$. From the total transfer function $h(s)$ in Problem 15, we then find

$$h(s) = \frac{R_a}{1 + T_a s} + \frac{R_a}{1 + T_a s} = 2\frac{R_a}{1 + T_a s},$$

in other words, we have a single time constant.

The correct answer is *Choice D*.

	Problem 16: The time constant T_e of the electric subsystem is — which answer is ***correct***?	Choice
A:	The variation in u_a is too small to determine the time constant	
B:	$T_\mathrm{e} = 4\,\mathrm{ms}$	
C:	$T_\mathrm{e} = 2\,\mathrm{ms}$	
D:	$T_\mathrm{e} = 0.5\,\mathrm{ms}$	×

▲

In a stack of N_c cells, the stack transfer function $h_\mathrm{s}(s)$ will be

$$h_\mathrm{s}(s) \propto \frac{1}{(1+T_\mathrm{e}s)^{N_\mathrm{c}}}. \tag{61}$$

It is of interest to consider whether this "cascade" of N_c transfer functions, each of form $\frac{1}{1+T_\mathrm{e}s}$, results in a significant overall dynamic effect. For such an assessment, we consider the following exponential function, $\exp(\tau s)$:

$$\exp(\tau s) = \exp\left(n\frac{\tau}{n}s\right) = \left(\exp\left(\frac{\tau}{n}s\right)\right)^n. \tag{62}$$

With large n, τ/n is small, and we have

$$\exp\left(\frac{\tau}{n}s\right) \approx 1+\frac{\tau}{n}s$$
$$\Downarrow$$
$$\exp(\tau s) \approx \left(1+\frac{\tau}{n}s\right)^n. \tag{63}$$

For linear systems, a transfer function $\exp(-\tau s) = 1/\exp(\tau s)$ describes a time delay of size τ.

Problem 17. Electrodynamics of stack

Use these considerations (Eqs. 61–63) to find an approximation to the transfer function $h_\mathrm{s}(s)$.

	Problem 17: Which approximate expression for the transfer function from $i''(s)$ to $u_\mathrm{a}(s)$, denoted $h_\mathrm{s}(s)$, is ***correct***?	Choice
A:	$h_\mathrm{s}(s) \propto \exp(N_\mathrm{c}T_\mathrm{e}s)$	
B:	$h_\mathrm{s}(s) \propto \exp\left(\frac{T_\mathrm{e}}{N_\mathrm{c}}s\right)$	
C:	$h_\mathrm{s}(s) \propto \exp(-N_\mathrm{c}T_\mathrm{e}s)$	
D:	$h_\mathrm{s}(s) \propto \exp\left(-\frac{T_\mathrm{e}}{N_\mathrm{c}}s\right)$	

▲

Solution 17. Electrodynamics of stack

Use these considerations (Eqs. 61–63) to find an approximation to the transfer function $h_\mathrm{s}(s)$. From

$$\exp(\tau s) \approx \left(1+\frac{\tau}{n}s\right)^n$$

we find that

$$(1+T_\mathrm{e}s)^{N_\mathrm{c}} = \left(1+\frac{\tau}{N_\mathrm{c}}s\right)^{N_\mathrm{c}} \approx \exp(\tau s)$$

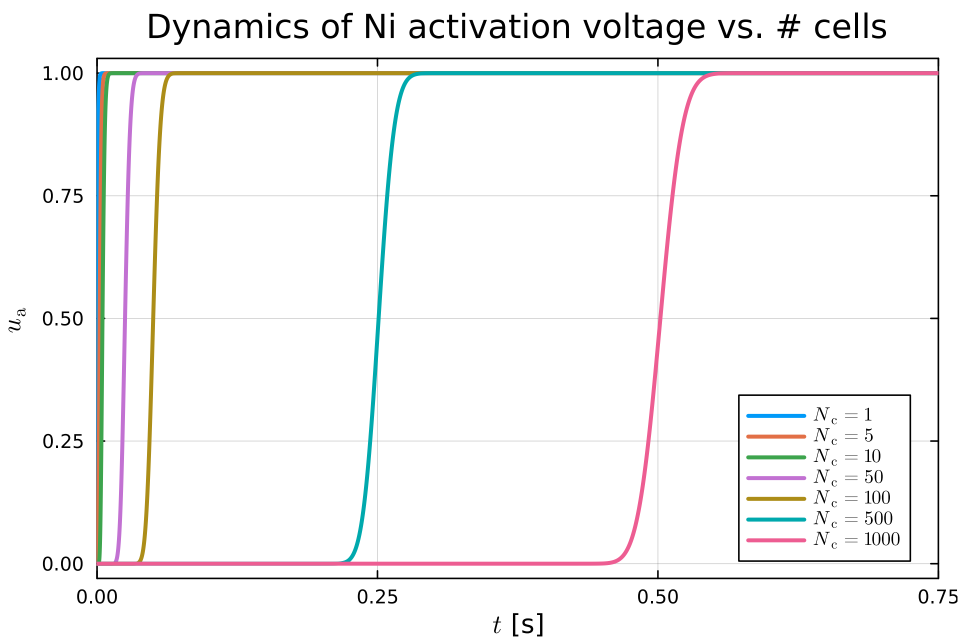

Figure 11: Case of N_c cascaded cells: total (normalized) activation voltage u_a after a step change in current density i''.

requires

$$T_\mathrm{e} = \frac{\tau}{N_\mathrm{c}} \Rightarrow \tau = N_\mathrm{c} T_\mathrm{e}.$$

Thus, for large N_c,

$$\left(1 + T_\mathrm{e} s\right)^{N_\mathrm{c}} \approx \exp\left(N_\mathrm{c} T_\mathrm{e} s\right).$$

It follows that the approximate transfer function for cascaded cells is

$$h_\mathrm{s}\left(s\right) = \frac{1}{\left(1 + T_\mathrm{e} s\right)^{N_\mathrm{c}}} \approx \frac{1}{\exp\left(N_\mathrm{c} T_\mathrm{e} s\right)} = \exp\left(-N_\mathrm{c} T_\mathrm{e} \cdot s\right).$$

This means that for a large number of cells, the electrodynamics gives approximately a time delay $\tau_\mathrm{s} \approx N_\mathrm{c} T_\mathrm{e}$. This means that *Choice C* is correct.

	Problem 17: Which approximate expression for the transfer function from $i''\left(s\right)$ to $u_\mathrm{a}\left(s\right)$, denoted $h_\mathrm{s}\left(s\right)$, is ***correct***?	Choice
A:	$h_\mathrm{s}\left(s\right) \propto \exp\left(N_\mathrm{c} T_\mathrm{e} s\right)$	
B:	$h_\mathrm{s}\left(s\right) \propto \exp\left(\frac{T_\mathrm{e}}{N_\mathrm{c}} s\right)$	
C:	$h_\mathrm{s}\left(s\right) \propto \exp\left(-N_\mathrm{c} T_\mathrm{e} s\right)$	×
D:	$h_\mathrm{s}\left(s\right) \propto \exp\left(-\frac{T_\mathrm{e}}{N_\mathrm{c}} s\right)$	

▲

Figure 11 shows the step-response in a normalized activation voltage $u_\mathrm{a} = u_\mathrm{a}^\mathrm{a} + u_\mathrm{a}^\mathrm{c}$ cascaded through N_c cells after a step change in i'', scaled to 1 V for numerical reasons. This provides a quantitative measure to assess the importance of including the electrodynamics in the PEM electrolyzer model.

Problem 18. Electrodynamics importance

Assess the importance of including electrodynamics in the model.

	Problem 18: Which of the following statements is ***correct***?	Choice
A:	With $T_e \approx 0.5\,\text{ms}$, the electric time constant is never relevant, and we can safely neglect the dynamics of the electric subsystem.	
B:	For unrealistically large values of N_c (N_c in the order of 1000), the electric time delay approaches the fast dynamics of gas pressures. Because there is considerable uncertainty in the electric time constant, the need to include an electric model should still be re-assessed depending on an improved activation voltage model (Butler-Volmer) and for the actual number of cells N_c in series.	
C:	Because the dynamics of the water levels are so slow, the electric time constant is never relevant, see Eq. 37.	
D:	There is no disadvantage in including the electric time constant, so we should always do that.	

▲

Solution 18. Electrodynamics importance

Figure 11 essentially confirms the result in Problem 17: although the electric time constant for a single cell may be small (e.g., $T_e \approx 0.5\,\text{ms}$), when coupled in series, a resulting time delay is multiplied up by N_c. With a realistic number of cells in series of, say, $N_c = 200$, the resulting time constant is still small, say $\tau_s = 0.1\,\text{s}$. However, as noted in connection with Fig. 10, depending on the parameters in the Butler-Volmer model, one may find an electric time constant T_e which is 10 times larger. And then the stack delay could approach $\tau_s = 1\,\text{s}$, even with $N_c = 200$. Observe also that the transfer function for the mass balance/oxygen gas dynamics in Eq. 36 has a time constant of $T = 2.4\,\text{s}$, which is very close to a time delay of $\tau_s = 1\,\text{s}$. It should also be observed that the slower time constants in the transfer functions in Eqs. 36, 37 really is the result of some (arbitrary) controller gains in the water level controllers; it might well be that these slow time constants in a real system would be much faster.

In summary, Choice A is not very convincing, as argued above. Choice C is not generally valid, since the dynamics of water levels depends on controller tuning. Choice D *may* sound correct, but there is always a disadvantage in including unnecessary fast (or slow) dynamics in that this dramatically increases the computation time and numeric complexity of the system. In summary, *Choice B* is correct.

	Problem 18: Which of the following statements is ***correct***?	Choice
A:	With $T_e \approx 0.5\,\text{ms}$, the electric time constant is never relevant, and we can safely neglect the dynamics of the electric subsystem.	
B:	For unrealistically large values of N_c (N_c in the order of 1000), the electric time delay approaches the fast dynamics of gas pressures. Because there is considerable uncertainty in the electric time constant, the need to include an electric model should still be re-assessed depending on an improved activation voltage model (Butler-Volmer) and for the actual number of cells N_c in series.	×
C:	Because the dynamics of the water levels are so slow, the electric time constant is never relevant, see Eq. 37.	
D:	There is no disadvantage in including the electric time constant, so we should always do that.	

▲

1.7 Electrolyzer efficiency

Three common measures of electrolyzer efficiency are:

LHV efficiency: The lower heating value (LHV) efficiency η_ℓ is defined as

$$\eta_\ell \triangleq \frac{u_\mathrm{n}}{u_\mathrm{c}} \tag{64}$$

where u_n is the equilibrium (Nernst) voltage $u_\mathrm{n} = \frac{\Delta_\mathrm{r}\tilde{G}}{\nu_{\mathrm{e}^-}F}$ at the operating temperature and pressures, and u_c is the cell voltage at the operating temperature, pressures, and current density i''.

HHV efficiency: The higher heating value (HHV) efficiency η_h is defined as

$$\eta_\mathrm{h} \triangleq \frac{u_\mathrm{th}}{u_\mathrm{c}} \tag{65}$$

where u_th is the thermoneutral voltage at the operating temperature and pressures, and u_c is the cell voltage at the operating temperature, pressures, and current density i''.

Production efficiency: The production efficiency is defined as the supplied electric power $\dot{W} = u_\mathrm{s} i_\mathrm{s} = N_\mathrm{c} u_\mathrm{c} i_\mathrm{c}$ per hydrogen production $\dot{m}_{\mathrm{H}_2}$,

$$\varepsilon_\mathrm{p} = \frac{\dot{W}}{\dot{m}_{\mathrm{H}_2}} = \frac{N_\mathrm{c} u_\mathrm{c} i_\mathrm{c}}{\dot{m}_{\mathrm{H}_2}} = \frac{N_\mathrm{c} A_\mathrm{c} u_\mathrm{c} i''}{\dot{m}_{\mathrm{H}_2}}. \tag{66}$$

Observe that contrary to η_ℓ and η_h, ε_p has dimension; ε_p is often given in units kW/ (kg/h) = kWh/kg.

Problem 19. Energy efficiency

Assume the following values taken from the group project: $\dot{m}_{\mathrm{H}_2}^{\max} = 24\,\mathrm{kg/h}$, $i''_{\max} = 2\,\mathrm{A/cm^2}$, $A_\mathrm{c} = 160\,\mathrm{cm^2}$, $N_\mathrm{c} = 2000$, $u_\mathrm{c} = 2.5\,\mathrm{V}$, $u_\mathrm{n} \approx 1.25\,\mathrm{V}$ (Fig. 4, $T = 60\,°\mathrm{C}$, $p_{\mathrm{H}_2} = 30\,\mathrm{bar}$), $u_\mathrm{th} \approx 1.49\,\mathrm{V}$ (Fig. 7). Find the three efficiencies η_ℓ, η_h, and ε_p.

	Problem 19: The energy efficiencies are — which statement is **correct**?	Choice
A:	$\eta_h = 0.4$, $\eta_\ell \approx 0.5$, $\varepsilon_\mathrm{p} \approx 50\,\mathrm{kWh/kg}$	
B:	$\eta_\ell = 0.4$, $\eta_\mathrm{h} \approx 0.5$, $\varepsilon_\mathrm{p} \approx 50\,\mathrm{kWh/kg}$	
C:	$\eta_h = 0.5$, $\eta_\ell \approx 0.6$, $\varepsilon_\mathrm{p} \approx 67\,\mathrm{kWh/kg}$	
D:	$\eta_\ell = 0.5$, $\eta_\mathrm{h} \approx 0.6$, $\varepsilon_\mathrm{p} \approx 67\,\mathrm{kWh/kg}$	

▲

Solution 19. Energy efficiency

Assume the following values taken from the group project: $\dot{m}_{\mathrm{H}_2}^{\max} = 24\,\mathrm{kg/h}$, $i''_{\max} = 2\,\mathrm{A/cm^2}$, $A_\mathrm{c} = 160\,\mathrm{cm^2}$, $N_\mathrm{c} = 2000$, $u_\mathrm{c} = 2.5\,\mathrm{V}$, $u_\mathrm{n} \approx 1.25\,\mathrm{V}$ (Fig. 4, $T = 60\,°\mathrm{C}$, $p_{\mathrm{H}_2} = 30\,\mathrm{bar}$), $u_\mathrm{th} \approx 1.49\,\mathrm{V}$ (Fig. 7). We find

$$\eta_\mathrm{h} = \frac{u_\mathrm{th}}{u_\mathrm{c}} = \frac{1.49}{2.5} \approx 0.6$$
$$\eta_\ell = \frac{u_\mathrm{n}}{u_\mathrm{c}} = \frac{1.25}{2.5} \approx 0.5.$$

Next,

$$\varepsilon_\mathrm{p} = \frac{N_\mathrm{c} A_\mathrm{c} u_\mathrm{c} i''}{\dot{m}_{\mathrm{H}_2}} = \frac{2000 \cdot 160 \cdot 2.5 \cdot 2}{24} = 66.7 \cdot 10^3\,\mathrm{Wh/kg} \approx 67\,\mathrm{kWh/kg}.$$

In conclusions, *Choice D* is correct.

	Problem 19: The energy efficiencies are — which statement is **correct**?	Choice
A:	$\eta_h = 0.4$, $\eta_\ell \approx 0.5$, $\varepsilon_\mathrm{p} \approx 50\,\mathrm{kWh/kg}$	
B:	$\eta_\ell = 0.4$, $\eta_\mathrm{h} \approx 0.5$, $\varepsilon_\mathrm{p} \approx 50\,\mathrm{kWh/kg}$	
C:	$\eta_h = 0.5$, $\eta_\ell \approx 0.6$, $\varepsilon_\mathrm{p} \approx 67\,\mathrm{kWh/kg}$	
D:	$\eta_\ell = 0.5$, $\eta_\mathrm{h} \approx 0.6$, $\varepsilon_\mathrm{p} \approx 67\,\mathrm{kWh/kg}$	×

Hand calculation: Obviously, $\eta_\mathrm{h} > \eta_\ell$, so we can immediately rule out options A and C. The simplest way is then to estimate η_ℓ, which is

$$\eta_\ell = \frac{u_\mathrm{n}}{u_\mathrm{c}} \approx \frac{1.25}{2.5} = 0.5.$$

We are then left with option D. We can, of course, check the other stated values, i.e., η_h and ε_p, just to make sure. We find

$$\eta_\mathrm{h} = \frac{u_\mathrm{th}}{u_\mathrm{c}} \approx \frac{1.49}{2.5} \approx \frac{1.5}{2.5} = \frac{3}{5} = 0.6.$$

Again, this fits option D. Finally, we have

$$\varepsilon_\mathrm{p} = \frac{N_\mathrm{c} A_\mathrm{c} u_\mathrm{c} i''}{\dot{m}_{\mathrm{H}_2}} = \frac{2000 \cdot 160 \cdot 2.5 \cdot 2}{24} = 2 \cdot 10^3 \frac{16}{24} \cdot 10 \cdot 5$$
$$\Downarrow$$
$$\varepsilon_\mathrm{p} = 10^5 \cdot \frac{2}{3} = 0.67 \cdot 10^5\,\mathrm{W/kg/h} = 67\,\mathrm{kWh/kg}.$$

Again, option D. So all three numbers point to option D.

▲

Problem 20. Energy efficiency range

Consider the HHV efficiency, the high heating value efficiency η_h. What is the possible *range* of η_h at the operating conditions (given temperature and pressures) of the group project when the current density i'' is allowed to change?

	Problem 20: The range of the HHV efficiency with varying i'' is approximately — which statement is **correct**?	Choice
A:	$\eta_\mathrm{h} \in [0, 1]$	
B:	$\eta_\mathrm{h} \in [0, 1.2]$	
C:	$\eta_\mathrm{h} \in [-0.2, 1]$	
D:	$\eta_\mathrm{h} \in [0.2, 1]$	

▲

Solution 20. Energy efficiency range

Consider the HHV efficiency, the high heating value efficiency η_{h}. What is the possible *range* of η_{h} at the operating conditions (given temperature and pressures) of the group project when the current density i'' is allowed to change?

We have

$$\eta_{\mathrm{h}} = \frac{u_{\mathrm{th}}}{u_{\mathrm{c}}}$$

where $u_{\mathrm{t}} = u_{\mathrm{t}}(T, p_j)$ while $u_{\mathrm{c}}(T, p_j, i'')$. With the given combinations of temperature and partial pressures of hydrogen and oxygen, the thermoneutral voltage is $u_{\mathrm{th}} \approx 1.49\,\mathrm{V}$. The cell voltage is $\lim_{i''\to 0^+} u_{\mathrm{c}} = u_{\mathrm{n}} \approx 1.25\,\mathrm{V}$ at the given temperature and pressures. It is less clear what happens with u_{c} in the limit $\lim_{i''\to\infty} u_{\mathrm{c}}$, because the model will break down then. But if we trust the model, u_{c} should become very large (the resistive voltage drop may dominate for large current densities, and we haven't even included the so-called "mass-diffusion" voltage drop). Thus, for low current, we have

$$\lim_{i''\to 0^+} \eta_{\mathrm{h}} \to \frac{u_{\mathrm{th}}}{u_{\mathrm{n}}} = \frac{1.49}{1.25} \approx 1.2$$

while

$$\lim_{i''\to\infty} \eta_{\mathrm{h}} \to \frac{u_{\mathrm{th}}}{\lim_{i''\to\infty} u_{\mathrm{c}}} = \frac{1.49}{\text{large number}} \approx 0.$$

In conclusion, $\eta_{\mathrm{h}} \in [0, 1.2]$, and *Choice B* is correct.

	Problem 20: The range of the HHV efficiency with varying i'' is approximately — which statement is **correct**?	Choice
A:	$\eta_{\mathrm{h}} \in [0, 1]$	
B:	$\eta_{\mathrm{h}} \in [0, 1.2]$	×
C:	$\eta_{\mathrm{h}} \in [-0.2, 1]$	
D:	$\eta_{\mathrm{h}} \in [0.2, 1]$	

▲

— END OF EXAM —